BBQ SPECIAL

This VSAQ Section contains a few 1 Mark Qs (Q7 to Q 12)

VERY EASY & CUTE SECTION WITH MANY MINI ANSWERS

VSAQ

SECTION-B

VSAQ CHAPTERS

10 Q X 2 M= 20 MARKS

★★	1.	RELATIONS & FUNCTIONS (RF) (2 Marks)	96 - 97
★★★	2.	MATRICES (MATS) (2 Marks)	98 - 102
★★★	3.	DETERMINANTS (DETS) (2 Marks)	103 - 104
★★★	4.	VECTOR ALGEBRA (VA) (2 Marks)	105 - 108
★★★	5.	3D GEOMETRY (3D) (2 Marks)	109 - 111
★★★	6.	PROBABILITY (PROB) (2 Marks)	112 - 114
★★★	7.	CONTINUITY & DIFFERENTIABILITY (CD) (2 Marks)	115 - 117
★★	8.	APPLICATION OF DERIVATIVES(AOD) (2 Marks)	118 - 119
★★★	9.	INTEGRATION (IG) (2 Marks)	120 - 124
★★★	10.	DIFFERENTIAL EQUATIONS (DE) (2 Marks)	125 - 127
•		STAR QUESTIONS PLUS (SQP)	128 - 129

Super Scoring Pages

★★★SSP★★★

Page 98 to 105, 110, 112, 115, 116, 118, 119, 120 to 124, 125, 127

Total SS Pages : 21

Mahesh

2 Marks

1 RELATIONS & FUNCTIONS

JUST 2 PAGES

153. Let A = {1, 2, 3}, B = {4, 5, 6, 7} and let f = {(1, 4), (2, 5), (3, 6)} be a function from A to B. Show that f is one-one.

Sol: Given that A = {1, 2, 3}, B = {4, 5, 6, 7}

f : A→B is defined as f = {(1, 4), (2, 5), (3, 6)}

∴ f(1) = 4, f(2) = 5, f(3) = 6

Thus, distinct elements of A have distinct images in B under f. ∴ f is one-one.

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154. Prove that the function f : R → R, given by f(x) = 2x, is one-one and onto.

Sol: f : R → R is given by f(x) = 2x. Let x_1, $x_2 \in R$ (domain)

(i) $f(x_1) = f(x_2) \Rightarrow 2x_1 = 2x_2 \Rightarrow x_1 = x_2$. ∴ f is one-one.

(ii) Let f(x) = y ⇒ y = 2x ⇒ $x = \frac{y}{2}$ exists ∀ y ∈ R (the codomain).

Now, $f(x) = 2x \Rightarrow f\left(\frac{y}{2}\right) = 2\left(\frac{y}{2}\right) = y$. ∴ f is onto.

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155. Check the injectivity and surjectivity of the function f : N → N given by f(x) = x^3 **[BMP]**

Sol: f : N → N; f(x) = x^3

(i) $f(x_1) = f(x_2) \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1 = x_2$ ∴ f is injective.

(ii) For 2 ∈ N we can't find in x in N such that $f(x) = x^3 = 2$ ∴ f is not surjective

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156. Show that the function f : R → R given by f(x) = x^3 is injective.

Sol: f : R → R is given by f(x) = x^3. Let $x_1, x_2 \in R$ (domain)

Now $f(x_1) = f(x_2) \Rightarrow x_1^3 = x_2^3 \Rightarrow x_1 = x_2$. ∴ f is injective

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157. Check the injectivity and surjectivity of the function f : Z → Z given by f(x) = x^2

Sol: f : Z → Z; f(x) = x^2

(i) $f(x_1) = f(x_2) \Rightarrow x_1^2 = x_2^2 \Rightarrow x_1 = \pm x_2$ ∴ f is not injective.

(ii) For 2 ∈ Z we can't find in x in Z such that $f(x) = x^2 = 2$ ∴ f is not surjective

Mahesh

158. **Show that the function $f : R \to R$, defined as $f(x) = x^2$, is neither one-one nor onto.**

Sol: (i) Given $f : R \to R$ is defined as $f(x) = x^2$. Take elements $1, -1$ in the domain R.

Now $f(-1) = (-1)^2 = 1$; $f(1) = 1^2 = 1$

Thus, two different elements $-1, 1$ of the domain have same image.

So, f is **not one-one.**

(ii) $f(x)=x^2 \Rightarrow$ Range of f contains all positive reals.

Moreover, Range does not contain any negative reals.

Given codomain is R. $\therefore$ Range $\neq$ Codomain.

So, f is **not onto.**

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159. **Define a reflexive relation**

Sol: **Reflexive Relation:** A relation R on a set A is reflexive if $(a,a) \in R$ for all $a \in A$.

Ex : The relation equality (=) on reals.

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160. **Define symmetric relation**

Sol: **Symmetric Relation:** A relation R on a set A is symmetric if $(a,b) \in R \Rightarrow (b,a) \in R$

Ex: The relation 'parallel of' lines ($||$) in a plane.

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161. **Define transitive relation**

Sol: **Transitive Relation:** A relation R on a set A is transitive

if $(a,b) \in R$ and $(b,c) \in R \Rightarrow (a,c) \in R$

Ex: The relation greater than (>) on reals.

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162. **Define an equivalence relation**

Sol: **Equivalence Relation:**

A relation R on a set A is called an equivalence relation

if R is reflexive, symmetric and transitive.

Ex: The relation equality (=) on reals.

Mahesh

2 Marks

2 MATRICES

MOST FAVOURITE CHAPTER TO ALL

163. If a matrix has 8 elements, what are the possible orders it can have?

Sol: If a matrix has 8 elements, then possible orders of the matrix have '8 as product'.
Hence, possible orders are 1×8, 8×1, 4×2, 2×4

PQ **If a matrix has 24 elements, what are the possible orders it can have?**
What, if it has 13 elements? [Ans: (1×13) and (13×1)] [For Answer ref. QR Code on P.No.12

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164. Find X, if $Y=\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}$ and $2X+Y=\begin{bmatrix}1 & 0\\ -3 & 2\end{bmatrix}$

Sol: $2X+Y=\begin{bmatrix}1 & 0\\ -3 & 2\end{bmatrix} \Rightarrow 2X+\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}=\begin{bmatrix}1 & 0\\ -3 & 2\end{bmatrix}$

$\Rightarrow 2X=\begin{bmatrix}1 & 0\\ -3 & 2\end{bmatrix}-\begin{bmatrix}3 & 2\\ 1 & 4\end{bmatrix}=\begin{bmatrix}-2 & -2\\ -4 & -2\end{bmatrix} \Rightarrow X=\frac{1}{2}\begin{bmatrix}-2 & -2\\ -4 & -2\end{bmatrix}=\begin{bmatrix}-1 & -1\\ -2 & -1\end{bmatrix}$.

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165. Find X and Y, if $X+Y=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix}$ and $X-Y=\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}$

Sol: Given $X+Y=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix}$.....(1) ; $X-Y=\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}$.....(2)

$(1)+(2) \Rightarrow 2X=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix}+\begin{bmatrix}3 & 0\\ 0 & 3\end{bmatrix}=\begin{bmatrix}10 & 0\\ 2 & 8\end{bmatrix}$

$\Rightarrow X=\frac{1}{2}\begin{bmatrix}10 & 0\\ 2 & 8\end{bmatrix}=\begin{bmatrix}5 & 0\\ 1 & 4\end{bmatrix} \Rightarrow X+Y=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix} \Rightarrow Y=\begin{bmatrix}7 & 0\\ 2 & 5\end{bmatrix}-\begin{bmatrix}5 & 0\\ 1 & 4\end{bmatrix} \Rightarrow Y=\begin{bmatrix}2 & 0\\ 1 & 1\end{bmatrix}$

PQ **Find X and Y, if $X+Y=\begin{bmatrix}5 & 2\\ 0 & 9\end{bmatrix}$ and $X-Y=\begin{bmatrix}3 & 6\\ 0 & -1\end{bmatrix}$** $\left[A: X=\begin{bmatrix}4 & 4\\ 0 & 4\end{bmatrix}, Y=\begin{bmatrix}1 & -2\\ 0 & 5\end{bmatrix}\right]$
[For Answer ref. QR Code on P.No.128]

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66. Find x and y, if $2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$

ol: Given $2\begin{bmatrix}1 & 3\\ 0 & x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$

$\Rightarrow \begin{bmatrix}2 & 6\\ 0 & 2x\end{bmatrix}+\begin{bmatrix}y & 0\\ 1 & 2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix} \Rightarrow \begin{bmatrix}2+y & 6\\ 1 & 2x+2\end{bmatrix}=\begin{bmatrix}5 & 6\\ 1 & 8\end{bmatrix}$

Equating the corresponding elements, $2+y=5 \Rightarrow y=3$; $2x+2=8 \Rightarrow x=3$ $\therefore x=3, y=3$

Mahesh

167. If $x\begin{bmatrix}2\\3\end{bmatrix}+y\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}10\\5\end{bmatrix}$ find the values of x and y.

Sol: $x\begin{bmatrix}2\\3\end{bmatrix}+y\begin{bmatrix}-1\\1\end{bmatrix}=\begin{bmatrix}10\\5\end{bmatrix}\Rightarrow\begin{bmatrix}2x\\3x\end{bmatrix}+\begin{bmatrix}-y\\y\end{bmatrix}=\begin{bmatrix}10\\5\end{bmatrix}\Rightarrow\begin{bmatrix}2x-y\\3x+y\end{bmatrix}=\begin{bmatrix}10\\5\end{bmatrix}$

Equating the corresponding elements $2x-y=10$.....(1); $3x+y=5$.....(2)

(1) + (2) $\Rightarrow 5x=15\Rightarrow x=3$

(2)$\Rightarrow 3x+y=5\Rightarrow y=5-3x \Rightarrow y=5-3(3)\Rightarrow y=5-9 \Rightarrow y=-4$ $\therefore$ x=3, y= –4

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168. Find the values of x,y and z from $\begin{bmatrix}x+y+z\\x+z\\y+z\end{bmatrix}=\begin{bmatrix}9\\5\\7\end{bmatrix}$

Sol: Equating the corresponding elements, we get

$x+y+z=9$.....(1); $x+z=5$.....(2); $y+z=7$.....(3)

From (1) and (2), we have $y+5=9\Rightarrow y=4$

From (3), we have $4+z=7\Rightarrow z=3$

Now $x+z=5\Rightarrow x+3=5\Rightarrow x=2$. Thus, $x=2, y=4, z=3$

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169. Find the values of x,y and z from $\begin{bmatrix}x+y & 2\\5+z & xy\end{bmatrix}=\begin{bmatrix}6 & 2\\5 & 8\end{bmatrix}$ [BMP]

Sol: Given $\begin{bmatrix}x+y & 2\\5+z & xy\end{bmatrix}=\begin{bmatrix}6 & 2\\5 & 8\end{bmatrix}$

Equating the corresponding elements, we get $x+y=6$; $xy=8$; $5+z=5\Rightarrow z=0$

Now $(x-y)^2=(x+y)^2-4xy\Rightarrow(x-y)^2=6^2-4(8)=36-32=4$

$\Rightarrow(x-y)^2=4\Rightarrow(x-y)=\pm2\Rightarrow x-y=2$ or $x-y=-2$

When $x-y=2$ and $x+y=6$ we get $x=4, y=2$

When $x-y=-2$ and $x+y=6$ we get $x=2, y=4$

Thus, $x=4, y=2, z=0$ or $x=2, y=4, z=0$

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170. Compute $\begin{bmatrix}\cos^2x & \sin^2x\\\sin^2x & \cos^2x\end{bmatrix}+\begin{bmatrix}\sin^2x & \cos^2x\\\cos^2x & \sin^2x\end{bmatrix}$

Sol: $\begin{bmatrix}\cos^2x & \sin^2x\\\sin^2x & \cos^2x\end{bmatrix}+\begin{bmatrix}\sin^2x & \cos^2x\\\cos^2x & \sin^2x\end{bmatrix}=\begin{bmatrix}\cos^2x+\sin^2x & \sin^2x+\cos^2x\\\sin^2x+\cos^2x & \cos^2x+\sin^2x\end{bmatrix}=\begin{bmatrix}1 & 1\\1 & 1\end{bmatrix}$

Mahesh

171. Construct a 2×2 matrix, $A = [a_{ij}]$, whose elements are given by $a_{ij} = \frac{(i+j)^2}{2}$

Sol: A 2×2 matrix is given by $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$

Given that $a_{ij} = \frac{(i+j)^2}{2}$; $i, j = 1, 2$

$\therefore a_{11} = \frac{(1+1)^2}{2} = \frac{4}{2} = 2$; $a_{12} = \frac{(1+2)^2}{2} = \frac{9}{2}$;

$a_{21} = \frac{(2+1)^2}{2} = \frac{9}{2}$; $a_{22} = \frac{(2+2)^2}{2} = \frac{16}{2} = 8$

$\therefore A = \begin{bmatrix} 2 & 9/2 \\ 9/2 & 8 \end{bmatrix}$

PQ Construct a 2×2 matrix whose elements are $a_{ij} = \frac{i}{j}$ $\left[\text{Ans: } A = \begin{bmatrix} 1 & 1/2 \\ 2 & 1 \end{bmatrix}\right]$

PQ Construct a 3×2 matrix whose elements are $a_{ij} = \frac{1}{2}|i - 3j|$ $\left[\text{Ans: } A = \begin{bmatrix} 1 & 5/2 \\ 1/2 & 2 \\ 0 & 3/2 \end{bmatrix}\right]$

[For Answers ref. QR Code on P.No.128]

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172. Let $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$. Find AB

Sol: $AB = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}\begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix} = \begin{bmatrix} 2(1)+4(-2) & 2(3)+4(5) \\ 3(1)+2(-2) & 3(3)+2(5) \end{bmatrix} = \begin{bmatrix} 2-8 & 6+20 \\ 3-4 & 9+10 \end{bmatrix} = \begin{bmatrix} -6 & 26 \\ -1 & 19 \end{bmatrix}$

PQ Find AB, if $A = \begin{bmatrix} 0 & -1 \\ 0 & 2 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 5 \\ 0 & 0 \end{bmatrix}$ $\left[\text{Ans: } \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}\right]$

PQ Let $A = \begin{bmatrix} 2 & 4 \\ 3 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 3 \\ -2 & 5 \end{bmatrix}$. Find BA $\left[\text{Ans: } BA = \begin{bmatrix} 11 & 10 \\ 11 & 2 \end{bmatrix}\right]$

[For Answers ref. QR Code on P.No.128]

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173. Compute $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}[2 \quad 3 \quad 4]$

Sol: $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}[2 \quad 3 \quad 4] = \begin{bmatrix} 1(2) & 1(3) & 1(4) \\ 2(2) & 2(3) & 2(4) \\ 3(2) & 3(3) & 3(4) \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 4 & 6 & 8 \\ 6 & 9 & 12 \end{bmatrix}$

PQ Compute $\begin{bmatrix} 1 & -2 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{bmatrix}$ $\left[\text{Ans: } \begin{bmatrix} -3 & -4 & 1 \\ 8 & 13 & 9 \end{bmatrix}\right]$

PQ Compute $\begin{bmatrix} 3 & -1 & 3 \\ -1 & 0 & 2 \end{bmatrix}\begin{bmatrix} 2 & -3 \\ 1 & 0 \\ 3 & 1 \end{bmatrix}$ $\left[\text{Ans: } \begin{bmatrix} 14 & -6 \\ 4 & 5 \end{bmatrix}\right]$

[For Answers ref. QR Code on P.No.1

Mahesh

174. Find AB, if $A=\begin{bmatrix}6 & 9\\ 2 & 3\end{bmatrix}$ and $B=\begin{bmatrix}2 & 6 & 0\\ 7 & 9 & 8\end{bmatrix}$

Sol: $AB=\begin{bmatrix}6 & 9\\ 2 & 3\end{bmatrix}\begin{bmatrix}2 & 6 & 0\\ 7 & 9 & 8\end{bmatrix}=\begin{bmatrix}6(2)+9(7) & 6(6)+9(9) & 6(0)+9(8)\\ 2(2)+3(7) & 2(6)+3(9) & 2(0)+3(8)\end{bmatrix}$

$=\begin{bmatrix}12+63 & 36+81 & 0+72\\ 4+21 & 12+27 & 0+24\end{bmatrix}=\begin{bmatrix}75 & 117 & 72\\ 25 & 39 & 24\end{bmatrix}$

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175. If $A=\begin{bmatrix}\sin\alpha & \cos\alpha\\ -\cos\alpha & \sin\alpha\end{bmatrix}$ then verify that $A'A=I$

Sol: Given that $A=\begin{bmatrix}\sin\alpha & \cos\alpha\\ -\cos\alpha & \sin\alpha\end{bmatrix}$ $\quad\therefore A'=\begin{bmatrix}\sin\alpha & -\cos\alpha\\ \cos\alpha & \sin\alpha\end{bmatrix}$

Now, $A'A=\begin{bmatrix}\sin\alpha & -\cos\alpha\\ \cos\alpha & \sin\alpha\end{bmatrix}\begin{bmatrix}\sin\alpha & \cos\alpha\\ -\cos\alpha & \sin\alpha\end{bmatrix}$

$=\begin{bmatrix}\sin\alpha\sin\alpha+(-\cos\alpha)(-\cos\alpha) & \sin\alpha\cos\alpha+(-\cos\alpha)\sin\alpha\\ \sin\alpha\cos\alpha+\sin\alpha(-\cos\alpha) & \cos\alpha\cos\alpha+\sin\alpha\sin\alpha\end{bmatrix}$

$=\begin{bmatrix}\cos^2\alpha+\sin^2\alpha & \sin\alpha\cos\alpha-\sin\alpha\cos\alpha\\ \sin\alpha\cos\alpha-\sin\alpha\cos\alpha & \cos^2\alpha+\sin^2\alpha\end{bmatrix}=\begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}=I$

Thus, $A'A=I$.

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176. Find x, if $\begin{bmatrix}x & -5 & -1\end{bmatrix}\begin{bmatrix}1 & 0 & 2\\ 0 & 2 & 1\\ 2 & 0 & 3\end{bmatrix}\begin{bmatrix}x\\ 4\\ 1\end{bmatrix}=O$

Sol: Given that $\begin{bmatrix}x & -5 & -1\end{bmatrix}\begin{bmatrix}1 & 0 & 2\\ 0 & 2 & 1\\ 2 & 0 & 3\end{bmatrix}\begin{bmatrix}x\\ 4\\ 1\end{bmatrix}=O$

$\Rightarrow\begin{bmatrix}x+0-2 & 0-10+0 & 2x-5-3\end{bmatrix}\begin{bmatrix}x\\ 4\\ 1\end{bmatrix}=[0]$

$\Rightarrow\begin{bmatrix}x-2 & -10 & 2x-8\end{bmatrix}\begin{bmatrix}x\\ 4\\ 1\end{bmatrix}=[0]\Rightarrow[x(x-2)-40+2x-8]=[0]$

$\Rightarrow[x^2-2x-40+2x-8]=[0]\Rightarrow[x^2-48]=[0]\Rightarrow x^2-48=0$

$\Rightarrow x^2=48\Rightarrow x=\pm4\sqrt{3}$

PQ **For what values of x :** $\begin{bmatrix}1 & 2 & 1\end{bmatrix}\begin{bmatrix}1 & 2 & 0\\ 2 & 0 & 1\\ 1 & 0 & 2\end{bmatrix}\begin{bmatrix}0\\ 2\\ x\end{bmatrix}=O$ [Ans: x= –1]

[For Answer ref. QR Code on P.No.128]

Mahesh

177. Show that the matrix $A=\begin{bmatrix}1 & -1 & 5\\ -1 & 2 & 1\\ 5 & 1 & 3\end{bmatrix}$ is a symmetric matrix.

Sol: Given that $A=\begin{bmatrix}1 & -1 & 5\\ -1 & 2 & 1\\ 5 & 1 & 3\end{bmatrix}$

Now $A'=\begin{bmatrix}1 & -1 & 5\\ -1 & 2 & 1\\ 5 & 1 & 3\end{bmatrix}=A$ Hence, A is a symmetric matrix.

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178. Show that the matrix $A=\begin{bmatrix}0 & 1 & -1\\ -1 & 0 & 1\\ 1 & -1 & 0\end{bmatrix}$ is a skew symmetric matrix.

Sol: $A=\begin{bmatrix}0 & 1 & -1\\ -1 & 0 & 1\\ 1 & -1 & 0\end{bmatrix}$

Now $A'=\begin{bmatrix}0 & -1 & 1\\ 1 & 0 & -1\\ -1 & 1 & 0\end{bmatrix}=-\begin{bmatrix}0 & 1 & -1\\ -1 & 0 & 1\\ 1 & -1 & 0\end{bmatrix}=-A$ Hence, A is a skew symmetric matrix.

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179. For the matrix $A=\begin{bmatrix}1 & 5\\ 6 & 7\end{bmatrix}$, verify that $(A+A')$ is a symmetric matrix

Sol: Given that $A=\begin{bmatrix}1 & 5\\ 6 & 7\end{bmatrix}\Rightarrow A'=\begin{bmatrix}1 & 6\\ 5 & 7\end{bmatrix}$

$A+A'=\begin{bmatrix}1 & 5\\ 6 & 7\end{bmatrix}+\begin{bmatrix}1 & 6\\ 5 & 7\end{bmatrix}=\begin{bmatrix}2 & 11\\ 11 & 14\end{bmatrix}$

Also $[(A+A')]'=\begin{bmatrix}2 & 11\\ 11 & 14\end{bmatrix}=(A+A')$ $\therefore (A+A')$ is a symmetric matrix

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180. Compute $\begin{bmatrix}a^2+b^2 & b^2+c^2\\ a^2+c^2 & a^2+b^2\end{bmatrix}+\begin{bmatrix}2ab & 2bc\\ -2ac & -2ab\end{bmatrix}$

Sol: $\begin{bmatrix}a^2+b^2 & b^2+c^2\\ a^2+c^2 & a^2+b^2\end{bmatrix}+\begin{bmatrix}2ab & 2bc\\ -2ac & -2ab\end{bmatrix}=\begin{bmatrix}a^2+b^2+2ab & b^2+c^2+2bc\\ a^2+c^2-2ac & a^2+b^2-2ab\end{bmatrix}=\begin{bmatrix}(a+b)^2 & (b+c)^2\\ (a-c)^2 & (a-b)^2\end{bmatrix}$

Mahesh

2 Marks

3 DETERMINANTS

CUTE 8 PROBLEMS IN JUST 2 PAGES

181. Evaluate $\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix}$

Sol: We have $\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix} = x(x)-(x+1)(x-1) = x^2-(x^2-1) = x^2-x^2+1 = 1$

PQ Evaluate $\begin{vmatrix} x^2-x+1 & x-1 \\ x+1 & x+1 \end{vmatrix}$ [Ans: x^3-x^2+2]

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182. Evaluate $\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix}$

Sol: $\begin{vmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{vmatrix} = (\cos\theta)(\cos\theta)-(-\sin\theta)(\sin\theta) = \cos^2\theta+\sin^2\theta = 1$

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183. Find the values of x for which $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$

Sol: We have $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$ $\left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad-bc\right]$

$\Rightarrow 3-x^2 = 3-8 \Rightarrow x^2 = 8 \Rightarrow x = \pm\sqrt{8} = \pm\sqrt{4\times2} = \pm 2\sqrt{2}$

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184. Find values of x, if $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$

Sol: Given that $\begin{vmatrix} 2 & 3 \\ 4 & 5 \end{vmatrix} = \begin{vmatrix} x & 3 \\ 2x & 5 \end{vmatrix}$

$\Rightarrow 2\times5-3\times4 = x\times5-3\times2x \Rightarrow 10-12 = 5x-6x \Rightarrow -2 = -x \Rightarrow x = 2$

PQ Find the values of x, if $\begin{vmatrix} 2 & 4 \\ 5 & 1 \end{vmatrix} = \begin{vmatrix} 2x & 4 \\ 6 & x \end{vmatrix}$ [Ans: $x = \pm\sqrt{3}$]

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185. If $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A| = 4|A|$

Sol: The given matrix is $A = \begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$

$\therefore 2A = 2\begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 8 & 4 \end{bmatrix}$ $\left[\because \begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad-bc\right]$

Vinesh

L.H.S $= |2A| = \begin{vmatrix} 2 & 4 \\ 8 & 4 \end{vmatrix} = 2\times4 - 4\times8 = 8 - 32 = -24$

Now, $|A| = \begin{vmatrix} 1 & 2 \\ 4 & 2 \end{vmatrix} = 1\times2 - 2\times4 = 2 - 8 = -6$ $\quad \therefore$ RHS $= 4|A| = 4(-6) = -24$

$\therefore |2A| = 4|A|$

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186. Write Minors and Cofactors of the elements of $\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$

Sol: The given determinant is $\begin{vmatrix} 2 & -4 \\ 0 & 3 \end{vmatrix}$

Minor of the element a_{ij} is M_{ij}

M_{11}= Minor of the element $a_{11} = 3$; $\quad M_{12}$= Minor of the element $a_{12} = 0$;

M_{21} = Minor of the element $a_{21} = -4$; $\quad M_{22}$= Minor of the element $a_{22} = 2$;

Now, cofactor of a_{ij} is $A_{ij} = (-1)^{i+j} M_{ij}$

$A_{11} = (-1)^{1+1}(3) = 3$; $\quad A_{12} = (-1)^{1+2}(0) = 0$

$A_{21} = (-1)^{2+1}(-4) = 4$; $\quad A_{22} = (-1)^{2+2}(2) = 2$

PQ **Write Minors and Cofactors of the elements of** $\begin{vmatrix} a & c \\ b & d \end{vmatrix}$

[Ans: Minors: d,b,c,a; Cofactors: d,–b,–c,a]

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187. Find adjA for $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ **[BMP]**

Sol: We have $A_{11} = 4, A_{12} = -1, A_{21} = -3, A_{22} = 2$ $\quad \therefore$ adjA $= \begin{bmatrix} A_{11} & A_{21} \\ A_{12} & A_{22} \end{bmatrix} = \begin{bmatrix} 4 & -3 \\ -1 & 2 \end{bmatrix}$

PQ **Find the adjoint of the matrix** $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ $\quad \left[\text{Ans: adjA} = \begin{bmatrix} 4 & -2 \\ -3 & 1 \end{bmatrix}\right]$

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188. Find the inverse of the matrix $\begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$

Sol: Let $A = \begin{bmatrix} 2 & -2 \\ 4 & 3 \end{bmatrix}$. Then, $|A| = (2\times3) - (-2\times4) = 6 - (-8) = 14$

Now, $A_{11} = 3$; $\quad A_{12} = -4$ $\quad A_{21} = 2$; $\quad A_{22} = 2$

Hence, adjA $= \begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$ $\quad \therefore A^{-1} = \frac{1}{|A|}\text{adjA} = \frac{1}{14}\begin{bmatrix} 3 & 2 \\ -4 & 2 \end{bmatrix}$

PQ **Find the inverse of the matrix** $\begin{bmatrix} -1 & 5 \\ -3 & 2 \end{bmatrix}$ $\quad \left[\therefore A^{-1} = \frac{1}{13}\begin{bmatrix} 2 & -5 \\ 3 & -1 \end{bmatrix}\right]$

Mahesh

2 Marks

4 VECTOR ALGEBRA

PHYSICS Qs IN MATHS

189. **Classify the following as scalar and vector quantities i)force ii) distance [BMP]**

Sol: i) Force-vector ii) Distance-scalar

PQ **Classify the following as scalar and vector quantities.**

i) work done ii) displacement [Ans: i) Work done - scalar.ii) displacement - vector]

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190. **Compute the magnitude of the vector $\vec{a}=\hat{i}+\hat{j}+\hat{k}$** [BMP]

Sol: Given that $\vec{a}=\hat{i}+\hat{j}+\hat{k}$ $\therefore |\vec{a}| = \sqrt{x^2+y^2+z^2} = \sqrt{1+1+1} = \sqrt{3}$

PQ Compute the magnitude of the vector $\vec{b}=2\hat{i}-7\hat{j}-3\hat{k}$

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191. **Find the sum of the vectors $\vec{a}=\hat{i}-2\hat{j}+\hat{k}$, $\vec{b}=-2\hat{i}+4\hat{j}+5\hat{k}$ and $\vec{c}=\hat{i}-6\hat{j}-7\hat{k}$.**

Sol: Given vectors $\vec{a}=\hat{i}-2\hat{j}+\hat{k}$, $\vec{b}=-2\hat{i}+4\hat{j}+5\hat{k}$ and $\vec{c}=\hat{i}-6\hat{j}-7\hat{k}$.

$\therefore \vec{a}+\vec{b}+\vec{c} = (1-2+1)\hat{i}+(-2+4-6)\hat{j}+(1+5-7)\hat{k} = 0\hat{i}-4\hat{j}-\hat{k} = -4\hat{j}-\hat{k}$

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192. **Find unit vector in the direction of vector $\vec{a}=2\hat{i}+3\hat{j}+\hat{k}$**

Sol: Given $\vec{a}=2\hat{i}+3\hat{j}+\hat{k} \Rightarrow |\vec{a}| = \sqrt{2^2+3^2+1^2} = \sqrt{14}$

$\therefore$ Unit vector $\hat{a} = \frac{1}{|\vec{a}|}\vec{a} = \frac{1}{\sqrt{14}}(2\hat{i}+3\hat{j}+\hat{k}) = \frac{2}{\sqrt{14}}\hat{i}+\frac{3}{\sqrt{14}}\hat{j}+\frac{1}{\sqrt{14}}\hat{k}$

PQ **Find the unit vector in the direction of the vector $\vec{a}=\hat{i}+\hat{j}+2\hat{k}$.** $\left[\text{A}: \frac{1}{\sqrt{6}}\hat{i}+\frac{1}{\sqrt{6}}\hat{j}+\frac{2}{\sqrt{6}}\hat{k}\right]$.

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193. **Find the unit vector in the direction of the sum of the vectors, $\vec{a}=2\hat{i}+2\hat{j}-5\hat{k}$ and $\vec{b}=2\hat{i}+\hat{j}+3\hat{k}$.**

Sol: Sum of the given vectors is $\vec{a}+\vec{b}\,(=\vec{c}, \text{say}) = 4\hat{i}+3\hat{j}-2\hat{k}$ and $|\vec{c}| = \sqrt{4^2+3^2+(-2)^2} = \sqrt{29}$

Thus, the required unit vector is $\hat{c} = \frac{1}{|\vec{c}|}\vec{c} = \frac{1}{\sqrt{29}}(4\hat{i}+3\hat{j}-2\hat{k}) = \frac{4}{\sqrt{29}}\hat{i}+\frac{3}{\sqrt{29}}\hat{j}-\frac{2}{\sqrt{29}}\hat{k}$

PQ **For given vectors, $\vec{a}=2\hat{i}-\hat{j}+2\hat{k}$ and $\vec{b}=-\hat{i}+\hat{j}-\hat{k}$. Find the unit vector in the direction of $\vec{a}+\vec{b}$.** $\left[\text{Ans}: \frac{1}{\sqrt{2}}\hat{i}+\frac{1}{\sqrt{2}}\hat{k}\right]$

Mahesh

194. Find a vector in the direction of vector $5\hat{i}-\hat{j}+2\hat{k}$ which has magnitude 8 units.

Sol: Let $\vec{a}=5\hat{i}-\hat{j}+2\hat{k}$. $\therefore$ A vector in the direction of vector $\vec{a}$ which has magnitude 8 units

$$=8\hat{a}=8\frac{\vec{a}}{|\vec{a}|}=\frac{8(5\hat{i}-\hat{j}+2\hat{k})}{\sqrt{25+1+4}}=\frac{8}{\sqrt{30}}(5\hat{i}-\hat{j}+2\hat{k})=\frac{40}{\sqrt{30}}\hat{i}-\frac{8}{\sqrt{30}}\hat{j}+\frac{16}{\sqrt{30}}\hat{k}$$

PQ **Find a vector in the direction of vector $\vec{a}=\hat{i}-2\hat{j}$ that has magnitude 7 units.**

$$\left[\text{Ans}: \frac{7}{\sqrt{5}}\hat{i}-\frac{14}{\sqrt{5}}\hat{j}\right]$$

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195. Find the value of x for which $x(\hat{i}+\hat{j}+\hat{k})$ is a unit vector.

Sol: If $x(\hat{i}+\hat{j}+\hat{k})$ is a unit vector then

$$\left|x(\hat{i}+\hat{j}+\hat{k})\right|=1\Rightarrow\sqrt{x^2+x^2+x^2}=1\Rightarrow\sqrt{3x^2}=1\Rightarrow\sqrt{3}x=1\Rightarrow x=\pm\frac{1}{\sqrt{3}}$$

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196. Find $|\vec{x}|$, if for a unit vector $\vec{a}, (\vec{x}-\vec{a}).(\vec{x}+\vec{a})=12$

Sol: Given that $\vec{a}$ is a unit vector $\Rightarrow|\vec{a}|=1$ .....(i)

Also $(\vec{x}-\vec{a}).(\vec{x}+\vec{a})=12\Rightarrow\vec{x}.\vec{x}+\vec{x}.\vec{a}-\vec{a}.\vec{x}-\vec{a}.\vec{a}=12\Rightarrow|\vec{x}|^2-|\vec{a}|^2=12$

Putting $|\vec{a}|=1$ from (i), $|\vec{x}|^2-1=12\Rightarrow|\vec{x}|^2=13\Rightarrow|\vec{x}|=\sqrt{13}$.

PQ **If $\vec{a}$ is a unit vector and $(\vec{x}-\vec{a})\cdot(\vec{x}+\vec{a})=8$, then find $|\vec{x}|$.** **[Ans: $|\vec{x}|=3$]**

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197. Show that the vectors $2\hat{i}-3\hat{j}+4\hat{k}$ and $-4\hat{i}+6\hat{j}-8\hat{k}$ are collinear.

Sol: Let $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}$ and $\vec{b}=-4\hat{i}+6\hat{j}-8\hat{k}=-2(2\hat{i}-3\hat{j}+4\hat{k})=-2\vec{a}$

Here, $\vec{b}=-2\vec{a}=m\vec{a}$ where $m=-2<0$

$\therefore$ Vectors $\vec{a}$ and $\vec{b}$ are collinear (unlike because $m=-2<0$)

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198. Find the direction cosines of the vector $\hat{i}+2\hat{j}+3\hat{k}$.

Sol: The given vector is $\vec{a}=\hat{i}+2\hat{j}+3\hat{k}\Rightarrow|\vec{a}|=\sqrt{1^2+2^2+3^2}=\sqrt{24}$

$$\hat{a}=\frac{\vec{a}}{|\vec{a}|}=\frac{\hat{i}+2\hat{j}+3\hat{k}}{\sqrt{14}}=\frac{1}{\sqrt{14}}\hat{i}+\frac{2}{\sqrt{14}}\hat{j}+\frac{3}{\sqrt{14}}\hat{k}$$

We know that direction consines of a vectors $\vec{a}$ are coefficients of $\hat{i}, \hat{j}, \hat{k}$ in $\hat{a}$ $\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}$

$\therefore$ Dc's of the given vector $=\left(\frac{1}{\sqrt{14}}, \frac{2}{\sqrt{14}}, \frac{3}{\sqrt{14}}\right)$

Mahesh

199. **Show that the vector $\hat{i}+\hat{j}+\hat{k}$ is equally inclined to the axes OX, OY and OZ.**

Sol: Let $\vec{a}=\hat{i}+\hat{j}+\hat{k} \Rightarrow |\vec{a}|=\sqrt{1^2+1^2+1^2}=\sqrt{3}$

DCs of $\vec{a}=\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)\Rightarrow \cos\alpha=\frac{1}{\sqrt{3}},\cos\beta=\frac{1}{\sqrt{3}},\cos\gamma=\frac{1}{\sqrt{3}}$

Hence, the vector is equally inclined to OX, OY and OZ.

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200. **Show that the direction cosines of a vector equally inclined to the axes OX, OY and OZ are $\pm\left(\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}},\frac{1}{\sqrt{3}}\right)$**

Sol: Let the inclination of the equally inclined vector be α.

Then the DCs of the vectors are $\cos\alpha, \cos\alpha, \cos\alpha$

$\therefore \cos^2\alpha+\cos^2\alpha+\cos^2\alpha=1\Rightarrow 3\cos^2\alpha=1\Rightarrow\cos^2\alpha=\frac{1}{3}\Rightarrow\cos\alpha=\pm\frac{1}{\sqrt{3}}$

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201. **Show that the vectors $2\hat{i}-3\hat{j}+4\hat{k}$ and $-4\hat{i}+6\hat{j}-8\hat{k}$ are collinear.**

Sol: Let $\vec{a}=2\hat{i}-3\hat{j}+4\hat{k}$ and $\vec{b}=-4\hat{i}+6\hat{j}-8\hat{k}=-2(2\hat{i}-3\hat{j}+4\hat{k})=-2\vec{a}$

Here, $\vec{b}=-2\vec{a}=m\vec{a}$ where $m=-2<0$

$\therefore$ Vectors $\vec{a}$ and $\vec{b}$ are collinear (unlike because $m=-2<0$)

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202. **Show that the points A(1, 2, 7), B(2, 6, 3) and C(3, 10, –1) are collinear.**

Sol: Given points are A(1, 2, 7), B(2, 6, 3) , C(3, 10, –1)

D.r's of line joining A and B = (1–2, 2–6, 7–3) = (–1,–4, 4)

D.r's of line joining A and C = (1–3, 2–10, 7+1) = (–2,–8, 8)= 2 (–1,–4, 4)

Hence D.r's of AB and AC are proportional. $\therefore$ A,B,C are collinear.

PQ **Show that the points $A(-2\hat{i}+3\hat{j}+5\hat{k})$, $B(\hat{i}+2\hat{j}+3\hat{k})$ and $C(7\hat{i}-\hat{k})$ are collinear.**

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203. **If $\vec{a}=5\hat{i}-\hat{j}-3\hat{k}$ and $\vec{b}=\hat{i}+3\hat{j}-5\hat{k}$, then show that the vectors $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ are perpendicular.**

Sol: $\vec{a}+\vec{b}=(5\hat{i}-\hat{j}-3\hat{k})+(\hat{i}+3\hat{j}-5\hat{k})=6\hat{i}+2\hat{j}-8\hat{k}$

$\vec{a}-\vec{b}=(5\hat{i}-\hat{j}-3\hat{k})-(\hat{i}+3\hat{j}-5\hat{k})=4\hat{i}-4\hat{j}+2\hat{k}$

Now $(\vec{a}+\vec{b}).(\vec{a}-\vec{b})=(6\hat{i}+2\hat{j}-8\hat{k}).(4\hat{i}-4\hat{j}+2\hat{k})=24-8-16=0$

Scalar product of $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$ is zero. Hence these are perpendicular vectors.

Manesh

204. If $\vec{a}=2\hat{i}+2\hat{j}+3\hat{k}$, $\vec{b}=-\hat{i}+2\hat{j}+\hat{k}$ and $\vec{c}=3\hat{i}+\hat{j}$ are such that $\vec{a}+\lambda\vec{b}$ is perpendicular to $\vec{c}$, then find the value of λ.

Sol: $\vec{a}+\lambda\vec{b}=(2\hat{i}+2\hat{j}+3\hat{k})+\lambda(-\hat{i}+2\hat{j}+\hat{k})=(2-\lambda)\hat{i}+(2+2\lambda)\hat{j}+(3+\lambda)\hat{k}$

Also $\vec{c}=3\hat{i}+\hat{j}=3\hat{i}+\hat{j}+0\hat{k}$

Given that $\vec{a}+\lambda\vec{b}$ is perpendicular to $\vec{c}\Rightarrow(\vec{a}+\lambda\vec{b}).\vec{c}=0$

$\Rightarrow(2-\lambda)3+(2+2\lambda)1+(3+\lambda)0=0$

$\Rightarrow 6-3\lambda+2+2\lambda=0\Rightarrow-\lambda+8=0\Rightarrow-\lambda=-8\Rightarrow\lambda=8$

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205. Find the projection of the vector $\vec{a}=2\hat{i}+3\hat{j}+2\hat{k}$ on the vector $\vec{b}=\hat{i}+2\hat{j}+\hat{k}$

Sol: The projection of vector $\vec{a}$ on the vector $\vec{b}$ is given by $\frac{1}{|\vec{b}|}(\vec{a}\cdot\vec{b})$

$$=\frac{(2\hat{i}+3\hat{j}+2\hat{k}).(\hat{i}+2\hat{j}+\hat{k})}{|\hat{i}+2\hat{j}+\hat{k}|}=\frac{(2\times1+3\times2+2\times1)}{\sqrt{(1)^2+(2)^2+(1)^2}}=\frac{10}{\sqrt{6}}=\frac{5}{3}\sqrt{6}$$

PQ **Find the projection of the vector $\hat{i}+3\hat{j}+7\hat{k}$ on the vector $7\hat{i}-\hat{j}+8\hat{k}$** [Ans: $\frac{60}{\sqrt{114}}$]

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206. Find $|\vec{a}\times\vec{b}|$, if $\vec{a}=2\hat{i}+\hat{j}+3\hat{k}$ and $\vec{b}=3\hat{i}+5\hat{j}-2\hat{k}$

Sol: We have $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\2&1&3\\3&5&-2\end{vmatrix}=\hat{i}(-2-15)-(-4-9)\hat{j}+(10-3)\hat{k}=-17\hat{i}+13\hat{j}+7\hat{k}$

$$\therefore|\vec{a}\times\vec{b}|=\sqrt{(-17)^2+(13)^2+(7)^2}=\sqrt{289+169+49}=\sqrt{507}$$

PQ **Find $|\vec{a}\times\vec{b}|$, if $\vec{a}=\hat{i}-7\hat{j}+7\hat{k}$ and $\vec{b}=3\hat{i}-2\hat{j}+2\hat{k}$** [Ans: $19\sqrt{2}$]

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207. Find the area of a parallelogram whose adjacent sides are given by the vectors $\vec{a}=3\hat{i}+\hat{j}+4\hat{k}$ and $\vec{b}=\hat{i}-\hat{j}+\hat{k}$

Sol: The area of a parallelogram with $\vec{a}$ and $\vec{b}$ as its adjacent sides is given by $|\vec{a}\times\vec{b}|$.

Now, $\vec{a}\times\vec{b}=\begin{vmatrix}\hat{i}&\hat{j}&\hat{k}\\3&1&4\\1&-1&1\end{vmatrix}=5\hat{i}+\hat{j}-4\hat{k}$

$\therefore|\vec{a}\times\vec{b}|=\sqrt{25+1+16}=\sqrt{42}$.

Thus, the required area is $\sqrt{42}$ sq.units

Vaheen

2 Marks

5 3D-GEOMETRY

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208. **If a line makes angles 90°, 135°, 45° with the x, y and z-axes respectively, find its direction cosines.**

Sol: Let l, m, n be the direction cosines of the given line.

Then $l = \cos 90^\circ = 0$; $m = \cos 135^\circ = -\frac{1}{\sqrt{2}}$; $n = \cos 45^\circ = \frac{1}{\sqrt{2}}$ $\left[\because \cos(90^\circ + 45^\circ) = -\sin 45^\circ\right]$

$\therefore$ the DC's of the line are $0, \frac{-1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$

PQ **If a line makes angle 90°, 60° and 30° with the positive direction of x, y and z-axis respectively, find its direction cosines.** [Ans : $0, 1/2, \frac{\sqrt{3}}{2}$]

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209. **If a line makes an angles of 60° and 45° with the direction of positive X and Z aixs respectively. Find the angle made by the line with the Y-aixs.**

Sol: Let l, m , n be the direction cosines of the given line.

The line makes an angle β with y-axis.

Then $l = \cos 60^\circ = \frac{1}{2}$; $m = \cos\beta$; $n = \cos 45^\circ = \frac{1}{\sqrt{2}}$

We know that $l^2 + m^2 + n^2 = 1$

$$\Rightarrow \frac{1}{4} + m^2 + \frac{1}{2} = 1 \Rightarrow m^2 + \frac{1+2}{4} = 1 \Rightarrow m^2 = 1 - \frac{3}{4} = \frac{1}{4} \Rightarrow \cos^2\beta = \frac{1}{4} \Rightarrow \cos\beta = \pm\frac{1}{2}$$

$\therefore \beta = 60^\circ$ (or) 120°

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210. **Find the direction cosines of a line which makes equal angles with the coordinate axes.**

Sol: Let the line makes an angle α with each of the coordinates axes.

Then $l = \cos\alpha$, $m = \cos\alpha$, $n = \cos\alpha$

We know that $l^2 + m^2 + n^2 = 1 \Rightarrow \cos^2\alpha + \cos^2\alpha + \cos^2\alpha = 1 \Rightarrow 3\cos^2\alpha = 1$

$$\Rightarrow \cos^2\alpha = \frac{1}{3} \Rightarrow \cos\alpha = \pm\frac{1}{\sqrt{3}},$$

$\therefore$ D.C's of the line are $\pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}, \pm\frac{1}{\sqrt{3}}$

Manish

211. Find the direction cosines of the joining the points (–2, 6, 7) and (1, 2, –5).

Sol: Let $A(x_1,y_1,z_1)=(-2,6,7)$, $B(x_2,y_2,z_2)=(1,2,-5)$

D.r's of AB $=(a,b,c)=(x_2-x_1,y_2-y_1,z_2-z_1)=(1+2,\ 2-6,\ -5-7)=(3,-4,-12)$

$\therefore \sqrt{a^2+b^2+c^2}=\sqrt{9+16+144}=\sqrt{169}=13$

DC's of AB $=\left(\dfrac{a}{\sqrt{a^2+b^2+c^2}},\dfrac{b}{\sqrt{a^2+b^2+c^2}},\dfrac{c}{\sqrt{a^2+b^2+c^2}}\right)=\left(\dfrac{3}{13},\dfrac{-4}{13},\dfrac{-12}{13}\right)$

212. Show that the points (2, 3, 4), (– 1, – 2, 1), (5, 8, 7) are collinear.

Sol: Given points are A(2, 3, 4), B(– 1, – 2, 1), C(5, 8, 7).

D.r's of line joining A and B $=(-1-2,-2-3,1-4)=(-3,-5,-3)=(3,5,3)$

D.r's of line joining A and C $=(5-2,8-3,7-4)=(3,5,3)$

Hence D.r's of AB and AC are proportional. $\therefore$ A,B,C are collinear.

213. If the points (2, –1, 3), (4, *a*, 1) and (3, 1, *b*) are collinear, then find the ratio between *a* and *b*?

Sol: Let $A(x_1,y_1,z_1)=(2,-1,3)$, $B(x_2,y_2,z_2)=(4,a,1)$, $C(x_3,y_3,z_3)=(3,1,b)$ **[BMP]**

If A,B,C are collinear then $\dfrac{x_1-x_2}{x_2-x_3}=\dfrac{y_1-y_2}{y_2-y_3}=\dfrac{z_1-z_2}{z_2-z_3}$

$\Rightarrow \dfrac{2-4}{4-3}=\dfrac{-1-a}{a-1}=\dfrac{3-1}{1-b}\Rightarrow \dfrac{-2}{1}=\dfrac{-1-a}{a-1}=\dfrac{2}{1-b}$

$\Rightarrow \dfrac{-2}{1}=\dfrac{-1-a}{a-1}\Rightarrow 2a-2=1+a\Rightarrow 2a-a=1+2\Rightarrow a=3$

$\Rightarrow \dfrac{-2}{1}=\dfrac{2}{1-b}\Rightarrow \dfrac{-1}{1}=\dfrac{1}{1-b}\Rightarrow -1+b=1\Rightarrow b=2$

$\therefore$ Required ratio $\dfrac{a}{b}=\dfrac{3}{2}\Rightarrow 3:2$

214. Find the equation of the line which passes through the point (1, 2, 3) and is parallel to the vector $3\hat{i}+2\hat{j}-2\hat{k}$.

Sol: Cartesian form with $(x_1,y_1,z_1)=(1,2,3)$ and the direction ratios (a, b, c)= (3,2,–2) is

$\dfrac{x-x_1}{a}=\dfrac{y-y_1}{b}=\dfrac{z-z_1}{c}\Rightarrow \dfrac{x-1}{3}=\dfrac{y-2}{2}=\dfrac{z-3}{-2}$

215. Find the angle between the following pairs of lines:
$\vec{r}=2\hat{i}-5\hat{j}+\hat{k}+\lambda(3\hat{i}+2\hat{j}+6\hat{k})$ and $\vec{r}=7\hat{i}-6\hat{k}+\mu(\hat{i}+2\hat{j}+2\hat{k})$

Sol: Required angle is the angle between $\vec{b_1}=3\hat{i}+2\hat{j}+6\hat{k}$ and $\vec{b_2}=\hat{i}+2\hat{j}+2\hat{k}$

$|\vec{b_1}|=\sqrt{3^2+2^2+6^2}=\sqrt{49}=7$; $|\vec{b_2}|=\sqrt{1^2+2^2+2^2}=\sqrt{9}=3$

$\vec{b_1}.\vec{b_2}=(3i+2j+6k).(i+2j+2k)=3(1)+2(2)+6(2)=3+4+12=19$

$\therefore \cos\theta=\left|\dfrac{\vec{b_1}.\vec{b_2}}{|\vec{b_1}||\vec{b_2}|}\right|=\left|\dfrac{19}{7\times3}\right|=\dfrac{19}{21}\Rightarrow\theta=\cos^{-1}\left(\dfrac{19}{21}\right)$

Manesh

PQ **Find the angle between the pair of lines given by** $\vec{r}=3\hat{i}+2\hat{j}-4\hat{k}+\lambda(\hat{i}+2\hat{j}+2\hat{k})$ **and** $\vec{r}=5\hat{i}-2\hat{j}+\mu(3\hat{i}+2\hat{j}+6\hat{k})$ $\left[\text{Ans}:\theta=\cos^{-1}\left(\frac{19}{21}\right)\right]$

216. **Find the angle between the following pair of lines:**

$\frac{x-2}{2}=\frac{y-1}{5}=\frac{z+3}{-3}$ and $\frac{x+2}{-1}=\frac{y-4}{8}=\frac{z-5}{4}$

Sol: Reducing the pair of lines into vector form $\vec{b_1}=2\hat{i}+5\hat{j}-3\hat{k}$ and $\vec{b_2}=-\hat{i}+8\hat{j}+4\hat{k}$

$|\vec{b_1}|=\sqrt{(2)^2+(5)^2+(-3)^2}=\sqrt{38}$; $|\vec{b_2}|=\sqrt{(-1)^2+(8)^2+(4)^2}=\sqrt{81}=9$

$\vec{b_1}.\vec{b_2}=(2\hat{i}+5\hat{j}-3\hat{k}).(-\hat{i}+8\hat{j}+4\hat{k})=2(-1)+5\times8+(-3)4=-2+40-12=26$

$\therefore\cos\theta=\left|\frac{\vec{b_1}.\vec{b_2}}{|\vec{b_1}||\vec{b_2}|}\right|=\left|\frac{26}{\sqrt{38}\times9}\right|=\frac{26}{9\sqrt{38}}\Rightarrow\theta=\cos^{-1}\left(\frac{26}{9\sqrt{38}}\right)$

PQ **Find the angle between the pair of lines** $\frac{x+3}{3}=\frac{y-1}{5}=\frac{z+3}{4}$ and $\frac{x+1}{1}=\frac{y-4}{1}=\frac{z-5}{2}$ $\left[\text{Ans}:\theta=\cos^{-1}\left(\frac{8\sqrt{3}}{15}\right)\right]$

217. **Show that the lines** $\frac{x-5}{7}=\frac{y+2}{-5}=\frac{z}{1}$ and $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$ **are perpendicular to each other.**

Sol: From the given lines we have $\vec{b_1}=7\hat{i}-5\hat{j}+\hat{k}$ and $\vec{b_2}=\hat{i}+2\hat{j}+3\hat{k}$

$\vec{b_1}.\vec{b_2}=(7\hat{i}-5\hat{j}+\hat{k}).(\hat{i}+2\hat{j}+3\hat{k})=7(1)-5\times2+1(3)=7-10+3=0$

$\therefore$ the given lines are perpendicular to each other.

PQ **Show that the pair of lines** $\frac{4-x}{3}=\frac{2y+1}{2}=\frac{z-1}{5}$ and $\frac{2x+1}{2}=\frac{3-y}{2}=\frac{3z-1}{3}$ **are perpendicular to each other.**

218. **If the lines** $\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y-1}{1}=\frac{z-6}{-5}$ **are perpendicular, find the value of k.**

Sol: From the given lines we have $\vec{b_1}=-3\hat{i}+2k\hat{j}+2\hat{k}$ and $\vec{b_2}=3k\hat{i}+\hat{j}-5\hat{k}$

When the lines are perpendicular then $\vec{b_1}.\vec{b_2}=0$

$\vec{b_1}.\vec{b_2}=(-3\hat{i}+2k\hat{j}+2\hat{k}).(3k\hat{i}+\hat{j}-5\hat{k})=0\Rightarrow-3(3k)+2k(1)+2(-5)=0$

$\Rightarrow-9k+2k-10=0\Rightarrow7k=-10\Rightarrow k=\frac{-10}{7}$

PQ **Find the values of p so that the lines**

$\frac{1-x}{3}=\frac{7y-14}{2p}=\frac{z-3}{2}$ and $\frac{7-7x}{3p}=\frac{y-5}{1}=\frac{6-z}{5}$ **are at right angles.** $\left[\text{Ans}:p=\frac{70}{11}\right]$

Mahesh

2 Marks

6 PROBABILITY

FIRST SIX ARE BEST SIX

219. If $P(A)=\frac{7}{13}$, $P(B)=\frac{9}{13}$ and $P(A\cap B)=\frac{4}{13}$, evaluate $P(A|B)$.

Sol: We have $P(A|B)=\frac{P(A\cap B)}{P(B)}=\frac{\frac{4}{13}}{\frac{9}{13}}=\frac{4}{\cancel{13}}\times\frac{\cancel{13}}{9}=\frac{4}{9}$

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220. Compute $P(A|B)$, if $P(B)=0.5$ and $P(A\cap B)=0.32$

Sol: Given that $P(B)=0.5$ and $P(A\cap B)=0.32$ $\therefore P(A/B)=\frac{P(A\cap B)}{P(B)}=\frac{0.32}{0.5}=\frac{32}{50}=\frac{16}{25}$

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221. If $P(A)=\frac{3}{5}$ and $P(B)=\frac{1}{5}$, find $P(A\cap B)$ if A and B are independent events.

Sol: Given that $P(A)=\frac{3}{5}$, $P(B)=\frac{1}{5}$. Also A,B are independent events.

$\therefore P(A\cap B)=P(A)P(B)=\frac{3}{5}\times\frac{1}{5}=\frac{3}{25}$

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222. A and B are two events such that $P(A)\neq 0$. Find $P(B|A)$, if [BMP]
(i) A is a subset of B (ii) $A\cap B=\phi$

Sol: (i) A is a subset of $B\Rightarrow A\cap B=A$. Then $P(A\cap B)=P(B\cap A)=P(A)$

$\therefore P(B|A)=\frac{P(B\cap A)}{P(A)}=\frac{P(A)}{P(A)}=1$

(ii) $A\cap B=\phi\Rightarrow P(A\cap B)=0$ $\therefore P(B|A)=\frac{P(A\cap B)}{P(A)}=\frac{0}{P(A)}=0,$ $\because P(A)\neq 0$

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223. Given that E and F are events such that $P(E)=0.6$, $P(F)=0.3$ and $P(E\cap F)=0.2$, find $P(E|F)$ and $P(F|E)$

Sol: Given that $P(E)=0.6$, $P(F)=0.3$, $P(E\cap F)=0.2$

$\therefore P(E/F)=\frac{P(E\cap F)}{P(F)}=\frac{0.2}{0.3}=\frac{2}{3}$ and $P(F/E)=\frac{P(F\cap E)}{P(E)}=\frac{0.2}{0.6}=\frac{1}{3}$.

Mahesh

224. Evaluate $P(A \cup B)$, if $2P(A) = P(B) = \frac{5}{13}$ and $P(A|B) = \frac{2}{5}$

Sol: Given that $2P(A) = P(B) = \frac{5}{13} \Rightarrow P(A) = \frac{5}{26}, P(B) = \frac{5}{13}$

Now $P(A|B) = \frac{2}{5} \Rightarrow \frac{P(A \cap B)}{P(B)} = \frac{2}{5} \Rightarrow P(A \cap B) = \frac{2}{5} P(B) = \frac{2}{\cancel{5}} \times \frac{\cancel{5}}{13} = \frac{2}{13}$

$\therefore P(A \cup B) = P(A) + P(B) - P(A \cap B) = \frac{5}{26} + \frac{5}{13} - \frac{2}{13} = \frac{5+10-4}{26} = \frac{11}{26}$.

---

225. Determine P(E|F) when a coin is tossed three times, where
E : at least two heads , F : at most two heads

Sol: **E : at least two heads** $\Rightarrow$ E={HHH, HHT, HTH, THH} $\Rightarrow$ n(E) = 4
F : at most two heads $\Rightarrow$ F={HHT, HTH, THH, HTT, THT, TTH, TTT} $\Rightarrow$ n(F) =7
Hence $E \cap F$ = {HHT, HTH, THH} $\Rightarrow$ $n(E \cap F) = 3$

Now $P(E) = \frac{4}{8} = \frac{1}{2}, P(F) = \frac{7}{8}$, $P(E \cap F) = \frac{n(E \cap F)}{n(S)} = \frac{3}{8}$

$\therefore P(E/F) = \frac{P(E \cap F)}{P(F)} = \frac{3/8}{7/8} = \frac{3}{\cancel{8}} \times \frac{\cancel{8}}{7} = \frac{3}{7}$.

PQ **Determine P(E|F) when a coin is tossed three times, where**
E : at most two tails , F : at least one tail. [Ans: 6/7]

---

226. An urn contains 10 black and 5 white balls. Two balls are drawn from the urn one after the other without replacement. What is the probability that both drawn balls are black?

Sol: Let A and B denote the events that first and second ball drawn are black.

The probability of getting a black ball in the first draw is $P(A) = \frac{10}{15}$

Probability of getting a black ball in the second drawn given first drawn is black $P(B|A) = \frac{9}{14}$

$P(A \cap B) = P(A)\, P(B|A) = \frac{10}{15} \times \frac{9}{14} = \frac{3}{7}$ ($\because$ first drawn ball is not replaced)

Mahesh

227. **Two cards are drawn at random in succession and without replacement from a pack of 52 playing cards. Find the probability that both the cards are black.**

Sol: There are 26 black cards in a deck of 52 cards.

Let A,B be the events of drawing black cards in the successive draws.

Probability of getting a black card in the first draw is $P(A)=\frac{26}{52}$

Probability of getting a black card in the second drawn given first drawn is black $P(B|A)=\frac{25}{51}$

$\therefore P(A\cap B)=P(A)P(B|A)=\frac{26}{52}\times\frac{25}{51}=\frac{25}{102}$ ($\because$ first drawn ball is not replaced)

---

228. **A family has two children. What is the probability that both the children are boys given that at least one of them is a boy ?**

Sol: Let b stand for boy and g for girl.

The sample space is S = {(b, b), (g, b), (b, g), (g, g)}

Let E and F denote the following events :

E : 'both the children are boys' then E = {(b,b)}

F : 'at least one of the child is a boy' then F = {(b,b), (g,b), (b,g)}

Now $E\cap F=\{(b,b)\}$. Thus $P(F)=\frac{3}{4}$ and $P(E\cap F)=\frac{1}{4}$

$\therefore P(E|F)=\frac{P(E\cap F)}{P(F)}=\frac{\frac{1}{4}}{\frac{3}{4}}=\frac{1}{\cancel{4}}\times\frac{\cancel{4}}{3}=\frac{1}{3}$

---

229. **If a leap year is selected at random, what is the chance that it will contain 53 tuesdays?**

Sol: In a leap year, there are 366 days i.e., 52 weeks and 2 days.

In 52 weeks, there are 52 Tuesdays.

$\therefore$ the probability that the leap year will contain 53 Tuesdays is equal to the probability that the remaining 2 days will be Tuesdays.

The remaining 2 days can be any of the following:

(Mon & Tue), (Tue & Wed), (Wed &Thu), (Thu & Fri), (Fri & Sat), (Sat & Sun) , (Sun & Mon)

Total number of cases =7; Favourable cases=2

Probability that a leap year will have 53 Tuesdays =2/7

Mahesh

2 Marks

7 CONTINUITY & DIFFERENTIABILITY

MANY STUDENTS LIKE DOING DIFFERENTIATION

230. Differentiate the function $\sin(x^2+5)$ with respect to x .

Sol: Let $y=\sin(x^2+5)$

$$\therefore \frac{dy}{dx}=\frac{d}{dx}\sin(x^2+5)=\cos(x^2+5)\frac{d}{dx}(x^2+5) \quad \left[\because \frac{d}{dx}\sin f(x)=\cos f(x)\frac{d}{dx}f(x)\right]$$

$$=\cos(x^2+5)(2x+0)=2x\cos(x^2+5). \quad \left[\because \frac{d}{dx}x^n=nx^{n-1} \text{ and } \frac{d}{dx}(c)=0\right]$$

---

231. Differentiate the function $\cos(\sqrt{x})$ with respect to x .

Sol: Let $y=\cos(\sqrt{x})$

$$\therefore \frac{dy}{dx}=\frac{d}{dx}\cos(\sqrt{x})=-\sin\sqrt{x}\frac{d}{dx}\sqrt{x}=-\sin\sqrt{x}\frac{1}{2\sqrt{x}} \quad \left[\because \frac{d}{dx}\sqrt{x}=\frac{1}{2\sqrt{x}}\right]$$

---

232. Differentiate the function $\cos(\sin x)$ with respect to x . **[BMP]**

Sol: Let $y=\cos(\sin x)$

$$\therefore \frac{dy}{dx}=\frac{d}{dx}\cos(\sin x)=-\sin(\sin x)\frac{d}{dx}(\sin x) \quad \left[\because \frac{d}{dx}\cos f(x)=-\sin f(x)\frac{d}{dx}f(x)\right]$$

$$=-\sin(\sin x)\cos x=-\cos x\sin(\sin x)$$

PQ Differentiate $\sin(\log x)$, $x>0$ w.r.t. x. $\left[\text{Ans}: \frac{\cos(\log x)}{x}\right]$

---

233. Differentiate $\log(\log x)$, $x>1$ w.r.t. x.

Sol: Let $y=\log(\log x)$

$$\therefore \frac{dy}{dx}=\frac{1}{\log x}\frac{d}{dx}(\log x)=\frac{1}{\log x}\frac{1}{x}=\frac{1}{x\log x}. \quad \left[\because \frac{d}{dx}\log f(x)=\frac{1}{f(x)}\frac{d}{dx}f(x)\right]$$

---

234. Differentiate e^{x^3} w.r.t. x.

Sol: Let $y=e^{x^3}=e^{(x^3)}$

$$\therefore \frac{dy}{dx}=e^{(x^3)}\frac{d}{dx}x^3=e^{(x^3)}3x^2=3x^2e^{(x^3)} \quad \left[\because \frac{d}{dx}e^{f(x)}=e^{f(x)}\frac{d}{dx}f(x)\right]$$

PQ Differentiate $e^{\sin^{-1}x}$ w.r.t. x . $\left[\text{Ans}: e^{\sin^{-1}x}\frac{1}{\sqrt{1-x^2}}\right]$

Mahesh

235. Find $\frac{dy}{dx}$ in $y=\sin^{-1}\left(\frac{2x}{1+x^2}\right)$

Sol: Put $x=\tan\theta$ then we have $\theta=\tan^{-1}x$

$$\therefore\ y=\sin^{-1}\left(\frac{2\tan\theta}{1+\tan^2\theta}\right)=\sin^{-1}(\sin 2\theta)=2\theta=2\tan^{-1}x$$

$$\therefore \frac{dy}{dx}=2\left(\frac{1}{1+x^2}\right)=\frac{2}{1+x^2}$$

PQ Find $\frac{dy}{dx}$ in $y=\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right), 0<x<1$ $\left[\text{Ans}: \frac{dy}{dx}=\frac{2}{1+x^2}\right]$

---

236. Find $\frac{dy}{dx}$ in $y=\tan^{-1}\left(\frac{3x-x^3}{1-3x^2}\right), -\frac{1}{\sqrt{3}}<x<\frac{1}{\sqrt{3}}$

Sol: Put $x=\tan\theta$ then we have $\theta=\tan^{-1}x$

$$\therefore y=\tan^{-1}\left(\frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}\right)=\tan^{-1}(\tan 3\theta)=3\theta=3\tan^{-1}x\ \therefore \frac{dy}{dx}=3\left(\frac{1}{1+x^2}\right)=\frac{3}{1+x^2}$$

---

237. Find $\frac{dy}{dx}$ of $x^2+xy+y^2=100$

Sol: Given that $x^2+xy+y^2=100 \Rightarrow \frac{d}{dx}x^2+\frac{d}{dx}xy+\frac{d}{dx}y^2=\frac{d}{dx}(100)$

$$\therefore\ 2x+\left(x\frac{d}{dx}y+y\frac{d}{dx}x\right)+2y\frac{dy}{dx}=0 \Rightarrow 2x+x\frac{dy}{dx}+y+2y\frac{dy}{dx}=0$$

$$\Rightarrow (x+2y)\frac{dy}{dx}=-2x-y \Rightarrow \frac{dy}{dx}=-\frac{(2x+y)}{x+2y}.$$

---

238. Find $\frac{dy}{dx}$, if $x=at^2$, $y=2at$.

Sol: Given that $x=at^2$, $y=2at$. Then $\frac{dx}{dt}=2at$ and $\frac{dy}{dt}=2a$

$$\therefore \frac{dy}{dx}=\frac{\frac{dy}{dt}}{\frac{dx}{dt}}=\frac{2a}{2at}=\frac{1}{t}$$

Mahesh

239. **Find $\frac{dy}{dx}$ of $x = a\cos\theta, y = b\cos\theta$**

Sol: Given that $x = a\cos\theta$ and $y = b\cos\theta$. Differentiating both eqns. w.r.t. θ have

$$\frac{dx}{d\theta} = \frac{d}{d\theta}(a\cos\theta) = a\frac{d}{d\theta}\cos\theta = -a\sin\theta \text{ and } \frac{dy}{d\theta} = \frac{d}{d\theta}(b\cos\theta) = b\frac{d}{d\theta}\cos\theta = -b\sin\theta$$

$$\therefore \frac{dy}{dx} = \frac{dy/d\theta}{dx/d\theta} = \frac{-b\sin\theta}{-a\sin\theta} = \frac{b}{a}.$$

PQ **Find $\frac{dy}{dx}$ of $x = a\sec\theta, y = b\tan\theta$** $\left[\text{Ans}: \frac{b}{a}\operatorname{cosec}\theta\right]$

---

240. **Find $\frac{dy}{dx}$, if $x = a(\theta + \sin\theta), y = a(1 - \cos\theta)$.**

Sol: Given that $x = a(\theta + \sin\theta)$ and $y = a(1 - \cos\theta)$. Then $\frac{dx}{d\theta} = a(1+\cos\theta), \frac{dy}{d\theta} = a(\sin\theta)$

$$\therefore \frac{dy}{dx} = \frac{\frac{dy}{d\theta}}{\frac{dx}{d\theta}} = \frac{a\sin\theta}{a(1+\cos\theta)} = \frac{2\sin\frac{\theta}{2}\cos\frac{\theta}{2}}{2\cos^2\frac{\theta}{2}} = \tan\frac{\theta}{2}$$

PQ **Find $\frac{dy}{dx}$ of $x = a(\theta - \sin\theta), y = a(1 + \cos\theta)$** $\left[\text{Ans}: -\cot\frac{\theta}{2}\right]$

---

241. **Find the second order derivative of $x^2 + 3x + 2$**

Sol: Let $y = x^2 + 3x + 2 \Rightarrow \frac{dy}{dx} = 2x + 3(1) + 0 = 2x + 3$. Again differentiating w.r.t x, we have

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{dy}{dx}\right) = 2(1) + 0 = 2$$

PQ **Find the second order derivative of x^{20}** $\left[\text{Ans}: 380x^{18}\right]$

---

242. **Find the second order derivative of log x**

Sol: Let $y = \log x \Rightarrow \frac{dy}{dx} = \frac{1}{x}$ Again differentiating w.r.t x, we have

$$\frac{d^2y}{dx^2} = \frac{d}{dx}\left(\frac{1}{x}\right) = \frac{d}{dx}(x^{-1}) = (-1)x^{-2} = \frac{-1}{x^2}.$$

---

243. **If $y = A\sin x + B\cos x$, then prove that $\frac{d^2y}{dx^2} + y = 0$**

Sol: Given that $y = A\sin x + B\cos x \Rightarrow \frac{dy}{dx} = A\cos x - B\sin x$ and

$$\frac{d^2y}{dx^2} = \frac{d}{dx}(A\cos x - B\sin x) = -A\sin x - B\cos x = -y \quad \Rightarrow \frac{d^2y}{dx^2} + y = 0$$

Mahesh

2 Marks

8 APPLICATIONS OF DERIVATIVES

ALL EASY APPLICATIONS IN JUST 2 PAGES

244. Show that the function given by f (x) = 3x + 17 is increasing on R.

Sol: Given f(x) =3x + 17⇒ f '(x)=3>0 ∀ x∈R. ∴f (x) is increasing on R.

PQ **Show that the function given by f (x) = 7x – 3 is increasing on R.**

245. Show that the function given by f (x) = e^{2x} is increasing on R.

Sol: Given f(x) = e^{2x} ⇒ f '(x)=e^{2x} (2)>0 ∀ x∈R. ∴ f (x) is increasing on R.

PQ **Prove that the logarithmic function is increasing on (0, ∞).**

246. Show that the function f given by f (x) = $x^3 - 3x^2 + 4x$, x ∈ R is increasing on R. **[BMP]**

Sol: Given that f (x) = $x^3 - 3x^2 + 4x$

⇒ f ′(x) = $3x^2 - 6x + 4 = 3(x^2 - 2x + 1) + 1 = 3(x - 1)^2 + 1 > 0$ ∀ x∈R

∴ f (x) is increasing on **R**.

PQ **Prove that the function given by f (x) = $x^3 - 3x^2 + 3x - 100$ is increasing in R.**

247. Prove that the function given by f (x) = cos x is decreasing in (0, π)

Sol: Given that f (x) = cos x ⇒ f ′(x) = – sin x<0 ∀ x∈(0, π) [∵ sinx>0 for x ∈(0, π)]

∴ f (x) is is decreasing in (0, π).

PQ **Show that the function given by f (x) = sin x isincreasing in** $\left(0, \frac{\pi}{2}\right)$

248. Find the intervals in which the function f given by f (x) = $x^2 - 4x + 6$ is

(a) increasing (b) decreasing

Sol: Given that f (x) = $x^2 - 4x + 6$ ⇒ f ′(x) = 2x – 4=2(x–2)

(i) f(x) is increasing when f'(x)>0 ⇒2(x–2)>0 ⇒x–2>0⇒x>2⇒x∈ (2, ∞)

(ii) f(x) is decreasing when f'(x)<0 ⇒2(x–2)<0 ⇒x<2⇒x∈ (– ∞, 2)

249. Find the intervals in which the function $x^2 + 2x - 5$ is strictly increasing or decreasing

Sol: Let f(x)= $x^2 + 2x - 5$ ⇒f '(x) = 2x + 2=2(x+1)

(i) f(x) is increasing when f '(x) > 0 ⇒ x + 1 > 0⇒ x > – 1⇒ x ∈ (–1, ∞)

(ii) f(x) is decreasing when f '(x) < 0⇒ x + 1 < 0 ⇒ x < – 1⇒ x ∈ (–∞, –1).

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250. **For what values of a the function f given by $f(x) = x^2 + ax + 1$ is increasing on [1, 2]?**

Sol: Given that $f(x) = x^2 + ax + 1 \Rightarrow f'(x) = 2x + a$.

Now, f has to be increasing on [1,2]. Also 2x+a is a linear function.

In [1,2] its minimum occurs at x=1 $\therefore$ minimum value =2(1)+a=2+a

Since, the function is increasing, $2 + a \geq 0 \Rightarrow a \geq -2$

PQ **Find the intervals in which the function f given by $f(x) = 2x^3 - 3x^2 - 36x + 7$ is (a) increasing (b) decreasing [A:** Increasing in $(-\infty,-2)\cup(3,\infty)$; decreasing in $(-2,3)$]

---

251. **Find the maximum and minimum values of $f(x) = 9x^2 + 12x + 2$**

Sol: Given that $f(x) = 9x^2 + 12x + 2 \Rightarrow f'(x) = 18x+12=6(3x+2)$

$\Rightarrow f''(x) = 18>0$ $\therefore$ f(x) has only minimum value but no maximum value.

At extreme values $f'(x)=0 \Rightarrow 3x+2=0 \Rightarrow 3x+2=0 \Rightarrow x=-\frac{2}{3}$

Minimum value of f is $f\left(-\frac{2}{3}\right)=9\left(\frac{-2}{3}\right)^2+12\left(-\frac{2}{3}\right)+2=9\left(\frac{4}{9}\right)-8+2=-2$

---

252. **Find the rate of change of the area of a circle per second with respect to its radius r when r = 5 cm.**

Sol: For the circle, we take radius = r and area =A

Area of circle $A = \pi r^2 \Rightarrow A = \pi r^2 \Rightarrow \frac{dA}{dr} = \pi(2r) = 2\pi r$ When $r = 5, \frac{dA}{dr} = 2\pi(5) = 10\pi.$ cm^2

Thus, the area of the circle is changing at the rate of 10π cm^2/s.

---

253. **Find the rate of change of the area of a circle with respect to its radius r when r = 3 cm** **[BMP]**

Sol: For the circle, we take radius = r and area =A Given r= 3 cm

Area of circle $A = \pi r^2 \Rightarrow A = \pi r^2 \Rightarrow \frac{dA}{dr} = \pi(2r) = 2\pi r$

When $r = 3, \frac{dA}{dr} = 2\pi(3) = 6\pi.$

Thus, the area of the circle is changing at the rate of 6π cm^2/s

---

254. **The radius of a circle is increasing uniformly at the rate of 3 cm/s. Find the rate at which the area of the circle is increasing when the radius is 10 cm.**

Sol: For the circle, we take radius = r and area =A. Area of circle $A=\pi r^2$.

Given $\frac{dr}{dt} = 3cm/s$ and r= 10 cm $\Rightarrow \frac{dA}{dt} = \frac{d}{dt}(\pi r^2) = 2\pi r\frac{dr}{dt} = 2\pi(10)(3) = 60\pi$ cm^2/s

PQ **The radius of an air bubble is increasing at the rate of 1/2 cm/s. At what rate is the volume of the bubble increasing when the radius is 1 cm? [A: 2π cm^3/s]**

Mahesh

2 Marks

9 INTEGRATION

ALL EASY INTEGRATION PROBLEMS ARE HERE FOR YOU

PROBLEMS ON INDEFINITE INTEGRALS

255. Find $\int(ax^2+bx+c)dx$

Sol: $\int(ax^2+bx+c)dx=\int ax^2\,dx+\int bx\,dx+\int c\,dx$

$=a\int x^2dx+b\int x^1dx+c\int 1dx \quad =a\frac{x^3}{3}+b\frac{x^2}{2}+cx+c_1$ where c_1 is the constant of integration.

---

256. Find integral of $\frac{2x}{1+x^2}$

Sol: Put $1+x^2=t \Rightarrow 2xdx=dt$

$\therefore\int\frac{2x}{1+x^2}dx=\int\frac{dt}{t}=\int\frac{1}{t}dt=\log|t|+c=\log|1+x^2|+c=\log(1+x^2)+c.$ $[\because t=1+x^2]$

---

257. Find the integral of $\frac{3x^2}{x^6+1}$

Sol: Put $x^3=t\Rightarrow 3x^2dx=dt$

$\therefore\int\frac{3x^2}{x^6+1}dx=\int\frac{dt}{t^2+1}=\tan^{-1}t+C=\tan^{-1}(x^3)+C$ $\left[\because\int\frac{1}{x^2+a^2}dx=\frac{1}{a}\tan^{-1}\frac{x}{a}+C\right]$

PQ Find integral of $\frac{x^3}{\sqrt{1-x^8}}$ $\left[\text{Hint}: x^4=t,\ \text{Ans}:\frac{1}{4}\sin^{-1}(x^4)+C\right]$

---

258. Find $\int\sec x(\sec x+\tan x)dx$

Sol: $\int\sec x(\sec x+\tan x)dx=\int(\sec^2x+\sec x\tan x)dx$

$=\int\sec^2xdx+\int\sec x\tan xdx \quad =\tan x+\sec x+c.$

PQ Find the $\int\text{cosecx}(\text{cosecx}+\cot x)dx$ **[BMP]** [Ans : $-\cot x-\text{cosecx}+C$]

Mahesh

259. **Find integral of** $\frac{\sin x}{(1+\cos x)^2}$

Sol: Put $1+\cos x=t \Rightarrow -\sin x dx=dt$

$$\therefore \int \frac{\sin x}{(1+\cos x)^2}dx = -\int \frac{dt}{t^2} = \frac{1}{t}+C = \frac{1}{(1+\cos x)}+C \qquad \left(\because \frac{d}{dx}\left(\frac{1}{x^2}\right)=\frac{-1}{x}\right)$$

PQ **Find integral of** $\frac{\sin x}{1+\cos x}$ **[Hint:** $1+\cos x=t$ **Ans:** $-\log|1+\cos x|+C$ **]**

PQ **Find integral of** $\frac{\cos x}{\sqrt{1+\sin x}}$ [Hint : $1+\sin x=t$, Ans : $2\sqrt{1+\sin x}+C$]

---

260. **Find integral of** $\frac{e^{\tan^{-1}x}}{1+x^2}$

Sol: Put $\tan^{-1}x=t \Rightarrow \frac{1}{1+x^2}dx=dt$ $\quad \therefore \int \frac{e^{\tan^{-1}x}}{1+x^2}dx = \int e^t dt = e^t+C = e^{\tan^{-1}x}+C$

PQ **Find integral of** $\frac{\sin^{-1}x}{\sqrt{1-x^2}}$ $\left[\text{Hint}: \sin^{-1}x=t,\ \text{Ans}: \frac{(\sin^{-1}x)^2}{2}+C\right]$

---

261. **Find integral of** $\frac{\cos\sqrt{x}}{\sqrt{x}}$

Sol: Put $\sqrt{x}=t \Rightarrow \frac{1}{2\sqrt{x}}dx=dt \Rightarrow \int \frac{\cos\sqrt{x}}{\sqrt{x}}dx = 2\int \cos t dt = 2\sin t+C = 2\sin\sqrt{x}+C$

---

262. **Find integral of** $\frac{(\log x)^2}{x}$ **[BMP]**

Sol: Put $\log x=t \Rightarrow \frac{1}{x}dx=dt \Rightarrow \frac{dx}{x}=dt$

$$\therefore \int \frac{(\log x)^2}{x}dx = \int (\log x)^2\left(\frac{dx}{x}\right) = \int t^2 dt = \frac{t^3}{3}+c = \frac{1}{3}(\log x)^3+c \qquad [\because t=\log x]$$

PQ **Find integral of** $\frac{(1+\log x)^2}{x}$

---

263. **Find the integral of** $e^x(\sin x+\cos x)$

Sol: We know that $\int e^x[f(x)+f'(x)]dx = e^x f(x)+C$. Here $f(x)=\sin x$ and $f'(x)=\cos x$

$$\therefore I = \int e^x(\sin x+\cos x)dx = e^x \sin x+C$$

Mahesh

264. Find the integral of $e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)$

Sol: We know that $\int e^x[f(x)+f'(x)]dx = e^x f(x)+C$; Here, $f(x)=\frac{1}{x}$ and $f'(x)=\frac{-1}{x^2}$

$$\therefore I=\int e^x\left(\frac{1}{x}-\frac{1}{x^2}\right)=e^x\left(\frac{1}{x}\right)+C=\frac{e^x}{x}+C$$

---

265. Find $\int \log x dx$

Sol: We take log x as the first function (u) and 1 as the second function(v) are intergal by parts

$$\int(\log x.1)dx = \log x\int 1dx-\int[\frac{d}{dx}(\log x)\int 1dx]dx = (\log x)\cdot x-\int\frac{1}{x}xdx = x\log x-x+C$$

---

266. Find the integral of x sin x

Sol: Let $I=\int x\sin x dx$ Taking u=x, v = sinx and integrating by parts, we have

$$I = x\int \sin x dx-\int\left[\left(\frac{d}{dx}(x)\right)\int \sin x dx\right]dx = x(-\cos x)-\int 1\cdot(-\cos x)dx$$

$$= -x\cos x+\sin x+C$$

---

267. Find the integral of $\frac{1}{x^2-9}$

Sol: $\int\frac{dx}{x^2-9}=\int\frac{dx}{x^2-3^2}=\frac{1}{2\times3}\log\left|\frac{x-3}{x+3}\right|+C=\frac{1}{6}\log\left|\frac{x-3}{x+3}\right|+C$

PQ Find $\int\frac{dx}{x^2-16}$ $\left[\text{Ans}:\frac{1}{8}\log\left|\frac{x-4}{x+4}\right|+C\right]$

---

268. Find the integral of $\frac{1}{\sqrt{9-25x^2}}$

Sol: Put $5x=t\Rightarrow 5dx=dt$ $\left[\because\int\frac{1}{\sqrt{a^2-x^2}}dx=\text{Sin}^{-1}\left(\frac{x}{a}\right)+C\right]$

$$\therefore\int\frac{1}{\sqrt{9-25x^2}}dx=\frac{1}{5}\int\frac{1}{\sqrt{9-t^2}}dt=\frac{1}{5}\int\frac{1}{\sqrt{3^2-t^2}}dt=\frac{1}{5}\sin^{-1}\left(\frac{t}{3}\right)+C=\frac{1}{5}\sin^{-1}\left(\frac{5x}{3}\right)+C$$

Mahesh

PROBLEMS ON DEFINITE INTEGRALS

269. Evaluate $\int_2^3 x^2 dx$

Sol: $I=\int_2^3 x^2 dx=\left[\frac{x^3}{3}\right]_2^3=\frac{3^3}{3}-\frac{2^3}{3}=9-\frac{8}{3}=\frac{27-8}{3}=\frac{19}{3}$

PQ Evaluate $\int_1^2 x^5 dx$ [Ans:21/2]

270. Evaluate $\int_4^5 e^x dx$

Sol: Let $I=\int_4^5 e^x dx=\left[e^x\right]_4^5=e^5-e^4=e^4(e-1)$

PQ Find $\int_0^1 e^x dx$ [Ans:e–1]

271. Evaluate $\int_0^1 \frac{dx}{1+x^2}$

Sol: Let $I=\int_0^1 \frac{dx}{1+x^2}=\left[\tan^{-1}x\right]_0^1=\tan^{-1}(1)-\tan^{-1}(0)=\frac{\pi}{4}$

272. Evaluate $\int_2^3 \frac{xdx}{x^2+1}$

Sol: Let $I=\int_2^3 \frac{x}{x^2+1}dx=\frac{1}{2}\int_2^3 \frac{2x}{x^2+1}dx=\frac{1}{2}\left[\log(1+x^2)\right]_2^3$

$=\frac{1}{2}[\log(1+3^2)-\log(1+2^2)]=\frac{1}{2}[\log(10)-\log(5)]=\frac{1}{2}\log\left(\frac{10}{5}\right)=\frac{1}{2}\log 2$

PQ Evaluate $\int_0^1 \frac{x}{x^2+1}dx$ $\left[\text{Ans}:\frac{1}{2}\log 2\right]$

273. Evaluate $\int_2^5 (x+2)^2\,dx$

Sol: $I=\int_2^5 (x+2)^2\,dx=\left[\frac{(x+2)^3}{3}\right]_2^5=\frac{1}{3}[(x+2)^3]_2^5=\frac{1}{3}[(5+2)^3-(2+2)^3]=\frac{1}{3}[7^3-4^3]=93$

274. Find $\int_0^{\pi/2} \sin^2 x\cos x\,dx$ [DI by Substitution]

Sol: Put $\sin x=t\Rightarrow \cos x\,dx=dt;\ x=0\Rightarrow t=\sin 0=0;\ x=\frac{\pi}{2}\Rightarrow t=\sin\frac{\pi}{2}=1$

$\therefore \int_0^{\pi/2} \sin^2 x\cos x\,dx=\int_0^1 t^2dt=\left[\frac{t^3}{3}\right]_0^1=\frac{1}{3}[1-0]=\frac{1}{3}$

Mahesh

275. Evaluate $\int_0^4 |x-1|\,dx$

Sol: $|x-1| = (x-1) \le 0$ when $0 \le x \le 1$ and $(x-1) \ge 0$ when $1 \le x \le 4$

$I = \int_0^4 |x-1|\,dx = \int_0^1 -(x-1)dx + \int_1^4 (x-1)dx \quad \left(\int_a^b f(x)dx = \int_b^c f(x)dx + \int_c^b f(x)dx\right)$

$= \left[x - \frac{x^2}{2}\right]_0^1 + \left[\frac{x^2}{2} - x\right]_1^4 = \left(1 - \frac{1}{2} - 0\right) + \left[(8-4) - \left(\frac{1}{2} - 1\right)\right] = \frac{1}{2} + \frac{9}{2} = 5$

[Ans:9]

PQ Evaluate $\int_2^8 |x-5|\,dx$.

---

276. Evaluate $\int_0^4 \frac{x^2}{1+x}\,dx$

Sol: $I = \int_0^4 \frac{x^2}{1+x}dx = \int_0^4 \frac{(x^2-1)+1}{1+x}\,dx = \int_0^4 \left(\frac{(x-1)\cancel{(x+1)}}{\cancel{1+x}} + \frac{1}{1+x}\right)dx = \int_0^4 (x-1)dx + \int_0^4 \frac{dx}{1+x}$

$= \left[\frac{x^2}{2} - x\right]_0^4 + [\log(1+x)]_0^4 = \frac{4^2}{2} - 4 + \log 5 - \log 1 - 0 = 4 + \log 5$

---

277. Evaluate $\int_0^{\frac{\pi}{2}} \cos^2 x\,dx$

Sol: Let $I = \int_0^{\frac{\pi}{2}} \cos^2 x\,dx = \int_0^{\pi/2} \left(\frac{1+\cos 2x}{2}\right)dx = \left[\frac{x}{2} + \frac{\sin 2x}{4}\right]_0^{\pi/2} = \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_0^{\pi/2}$

$= \frac{1}{2}\left[\left(\frac{\pi}{2} + \frac{\sin \pi}{2}\right) - \left(0 + \frac{\sin 0}{2}\right)\right] = \frac{1}{2}\left[\frac{\pi}{2} + 0 - 0 - 0\right] = \frac{\pi}{4}$

---

278. Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sin x}{1+\cos^2 x}dx$

Sol: Put $\cos x = t \Rightarrow -\sin x\,dx = dt$. When $x = 0$ we have $t = 1$ and when $x = \frac{\pi}{2}$ we have $t = 0$

$\Rightarrow \int_0^{\frac{\pi}{2}} \frac{\sin x}{1+\cos^2 x}dx = -\int_1^0 \frac{dt}{1+t^2} = -\left[\tan^{-1} t\right]_1^0 = -\left[\tan^{-1} 0 - \tan^{-1} 1\right] = -\left[-\frac{\pi}{4}\right] = \frac{\pi}{4}$

---

279. Evaluate $\int_0^{\frac{\pi}{4}} \sin 2x\,dx$

Sol: $\int_0^{\frac{\pi}{4}} \sin 2x\,dx = \left[\frac{-\cos 2x}{2}\right]_0^{\frac{\pi}{4}} = \left[\frac{-\cos 2\frac{\pi}{4}}{2}\right] - \left[\frac{-\cos 0}{2}\right] = 0 - \left(\frac{-1}{2}\right) = 0 + \frac{1}{2} = \frac{1}{2}. \quad \left[\because \cos\frac{\pi}{2} = 0\right]$

Mahesh

2 Marks

10 DIFFERENTIAL EQUATIONS

THIS FIRST PAGE IS BEST & PROBLEMS IN 3RD PAGE ARE EASY

280. **Find the order and degree, if** $xy\frac{d^2y}{dx^2}+x\left(\frac{dy}{dx}\right)^2-y\frac{dy}{dx}=0$

Sol: 'Highest order derivative' in the given D.E is $\frac{d^2y}{dx^2}$. ∴ Order of DE is 2.

Power of $\frac{d^2y}{dx^2}$ is 1. ∴ Degree of DE is 1.

PQ **Find the order and degree of** $\frac{d^2y}{dx^2}+5x\left(\frac{dy}{dx}\right)^2-6y=\log x$ **[A:** order is 2, degree is 1.]

---

281. **Determine order and degree of** $\left(\frac{ds}{dt}\right)^4+3s\frac{d^2s}{dt^2}=0$ **[BMP]**

Sol: 'Highest order derivative' in the given DE is $\frac{d^2s}{dt^2}$. ∴ Order of DE is 2.

Power of $\frac{d^2s}{dt^2}$ is 1. ∴ Degree of DE is 1.

PQ **Find the order and degree of** $\left(\frac{dy}{dx}\right)^3-4\left(\frac{dy}{dx}\right)^2+7y=\sin x$ **[A:** Order is 1. Degree is3.]

---

282. **Determine order and degree of** $y''+(y')^2+2y=0$

Sol: 'Highest order derivative' present in the given DE is y". ∴ Order of DE is 2.

Power of y" is one. ∴ Degree of DE is 1.

PQ **Determine order and degree of** $y'''+2y''+y'=0$ [Ans: Order is 3. Degree is 1.]

---

283. **Determine order and degree of** $(y''')^2+(y'')^3+(y')^4+y^5=0$

Sol: 'Highest order derivative' in the given D.E is y'''. ∴ Order of DE is 3.

Power of y''' is 2. ∴ Degree of DE is 1.

PQ **Determine order and degree of** $y''+2y'+\sin y=0$ **[Ans:** Order is two.Degree is 1]

Mahesh

284. Verify that y=Ax is a solution to $xy' = y,\ (x \neq 0)$

Sol: Given function is $y = Ax \Rightarrow y' = \frac{d}{dx}(Ax) = A$

Verification: L.H.S=$xy' = xA = Ax = y$ = R.H.S . Hence verified.

---

285. Verify that the function $y = a\cos x + b\sin x$, where, $a, b \in R$ is a solution of the differential equation $\frac{d^2y}{dx^2} + y = 0$

Sol: The given function is $y = a\cos x + b\sin x$ ... (1) $\Rightarrow \frac{dy}{dx} = -a\sin x + b\cos x$

$\frac{d^2y}{dx^2} = -a\cos x - b\sin x = -(a\cos x + b\sin x) = -y \Rightarrow \frac{d^2y}{dx^2} + y = 0$. Hence verified.

---

286. Verify that $y= x^2+2x+C$ is a solution to $y' - 2x - 2 = 0$

Sol: Given function is $y = x^2 + 2x + c \Rightarrow y' = \frac{d}{dx}(x^2 + 2x + c) \Rightarrow y' = 2x + 2$

Verification : $y' - 2x - 2 = (2x + 2) - 2x - 2 = 0$. Hence verified.

---

287. Verify that y=cosx+C is a solution to $y' + \sin x = 0$

Sol: Given function is $y = \cos x + C \Rightarrow y' = \frac{d}{dx}(\cos x + C) \Rightarrow y' = -\sin x$

Verification: $y' + \sin x = -\sin x + \sin x = 0$. Hence verified.

---

288. Verify that $y = \sqrt{1+x^2}$ is a solution to $y' = \frac{xy}{1+x^2}$

Sol: Given function is $y = \sqrt{1+x^2} \Rightarrow y' = \frac{d}{dx}\left(\sqrt{1+x^2}\right) = \frac{1}{2\sqrt{1+x^2}} \cdot \frac{d}{dx}(1+x^2)$

$= \frac{2x}{2\sqrt{1+x^2}} = \frac{x}{2\sqrt{1+x^2}} = \frac{x(\sqrt{1+x^2})}{(1+x^2)} = \frac{xy}{1+x^2}$

$\therefore y' = \frac{xy}{1+x^2}$ Hence satisfied and verified.

Mahesh

289. **Find the general solution of the differential equation** $\frac{dy}{dx}=\frac{1+y^2}{1+x^2}$

Sol: Given De is written as $\frac{dy}{1+y^2}=\frac{dx}{1+x^2}\Rightarrow\int\frac{dy}{1+y^2}=\int\frac{dx}{1+x^2}$

$\Rightarrow \tan^{-1}y=\tan^{-1}x+C$ .This is the general solution of given DE.

PQ **Find the general solution of** $\frac{dy}{dx}=(1+x^2)(1+y^2)$ $\left[\text{Ans}: x+\frac{x^3}{x}+C\right]$

---

290. **Find the general solution of the differential equation** $\frac{dy}{dx}+\sqrt{\frac{1-y^2}{1-x^2}}=0$

Sol: Given D.E is $\frac{dy}{dx}+\sqrt{\frac{1-y^2}{1-x^2}}=0\Rightarrow\frac{dy}{dx}=-\frac{\sqrt{1-y^2}}{\sqrt{1-x^2}}\Rightarrow\frac{dy}{\sqrt{1-y^2}}=-\frac{dx}{\sqrt{1-x^2}}$

Integrating both sides, we get $\sin^{-1}y=-\sin^{-1}x+C\Rightarrow\sin^{-1}x+\sin^{-1}y=C$

---

291. **Find the general solution of** $\frac{dy}{dx}+\frac{y}{x}=x^2$

Sol: Given D.E is in the form $\frac{dy}{dx}+Py=Q$. Here $P=\frac{1}{x}$ and $Q=x^2$

$IF=e^{\int Pdx}=e^{\int\frac{1}{x}dx}=e^{\log x}=x$

General Solution is $y(I.F)=\int(Q\times IF)dx+C$

$\therefore yx=\int(x^2.x)dx+C\Rightarrow yx=\int x^3dx+C\Rightarrow xy=\frac{x^4}{4}+C$

---

292. **Find a particular solution of** $\frac{dy}{dx}=y\tan x;\ y=1\ \text{when}\ x=0$

Sol: Given D.E is $\frac{dy}{dx}=y\tan x\Rightarrow\frac{dy}{y}=\tan x dx$

Integrating both sides, we get $\int\frac{dy}{y}=\int\tan x dx\Rightarrow\log y=\log(\sec x)+\log C$

$\Rightarrow\log y=\log(\sec xC)\Rightarrow y=C\sec x$

Now, $y=1, x=0\Rightarrow 1=C\times\sec 0\Rightarrow 1=C\times 1\Rightarrow C=1$, Thus y=secx is a particular solution.

Mahesh

PQ QR Code

VSAQ STAR QUESTIONS PLUS

SQP QR Code

MATRICES & DETERMINANTS

1. Solve the equation for x, y, z and t, if $2\begin{bmatrix} x & z \\ y & t \end{bmatrix}+3\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}=3\begin{bmatrix} 3 & 5 \\ 4 & 6 \end{bmatrix}$

2. Given , $3\begin{bmatrix} x & y \\ z & w \end{bmatrix}=\begin{bmatrix} x & 6 \\ -1 & 2w \end{bmatrix}+\begin{bmatrix} 4 & x+y \\ z+w & 3 \end{bmatrix}$ find the values of x, y, z and w.

3. Simplify $\cos\theta\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}+\sin\theta\begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$

4. For the matrix $A=\begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix}$, verify that $(A+A')$ is a symmetric matrix

5. For the matrix $A=\begin{bmatrix} 1 & 5 \\ 6 & 7 \end{bmatrix}$, verify that $(A-A')$ is a skew symmetric matrix

6. Find the minor of element 6 in the determinant $\Delta=\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

7. If $A'=\begin{bmatrix} -2 & 3 \\ 1 & 2 \end{bmatrix}$ and $B=\begin{bmatrix} -1 & 0 \\ 1 & 2 \end{bmatrix}$, then find $(A+2B)'$

CONTINUITY & DIFFERENTIABILITY

8. Find $\frac{dy}{dx}$ if $x-y=\pi$.

9. Find $\frac{dy}{dx}$, if $y+\sin y=\cos x$

10. Find $\frac{dy}{dx}$ of $2x+3y=\sin x$

11. Find $\frac{dy}{dx}$ of $2x+3y=\sin y$

12. Find $\frac{dy}{dx}$ of $\sin^2 x+\cos^2 y=1$

13. Find $\frac{dy}{dx}$ of $y=\sec^{-1}\left(\frac{1}{2x^2-1}\right), 0<x<\frac{1}{\sqrt{2}}$

Mahesh

14. Differentiate $\log(\cos e^x)$ w.r.t. x.

15. Differentiate $\cos(\log x + e^x)$, $x > 0$ w.r.t. x.

INTEGRALS

INDEFINITE INTEGRALS & DEFINITE INTEGRALS

16. Find integral of $\frac{1}{x + x\log x}$

17. Find integral of $\sqrt{ax+b}$

18. Find integral of e^{2x+3}

19. Find integral of $\frac{e^{2x}-1}{e^{2x}+1}$

20. Find integral of $\sqrt{\sin 2x}\cos 2x$

21. Find integral of $\frac{\sin x}{1+\cos x}$

22. Find $\int \frac{dx}{x^2-6x+13}$

23. Find the integral of $\frac{x^2}{1-x^6}$

24. Evaluate $\int_2^3 \frac{1}{x} dx$

25. Evaluate $\int_1^2 (4x^3 - 5x^2 + 6x + 9) dx$

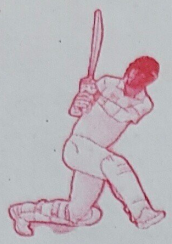
26. Evaluate $\int_0^{\frac{\pi}{4}} \tan x dx$

27. Evaluate $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

28. Evaluate $\int_2^3 \frac{dx}{x^2-1}$

29. Evaluate $\int_0^{\pi/2} \sin^3 x \, dx$

Mahesh



MULTIPLE CHOICE QUESTIONS

MCQ

SECTION-A

SCAN ABOVE FOR MCQ SOLUTIONS

TARGET MARKS: ≥5

MCQ CHAPTERS

6 MARKS

1.	RELATIONS & FUNCTIONS	131
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- MCQ Chapters & OMQ Chapters may be overlapped in IPE.
- OMQ & VSAQ may also be overlapped.

Mahesh

1) RELATIONS & FUNCTIONS

I. Select the correct option from the given choices.

1. Let R be the relation in the set {1, 2, 3, 4} given by [BMP] [2]
$R = \{(1, 2), (2, 2), (1, 1), (4,4), (1, 3), (3, 3), (3, 2)\}$. Choose the correct answer.

1) R is reflexive and symmetric but not transitive.

✓**2) R is reflexive and transitive but not symmetric.**

3) R is symmetric and transitive but not reflexive. 4)R is an equivalence relation.

2. Let R be the relation in the set N given by $R = \{(a, b) : a = b - 2, b > 6\}$. [3]
Choose the correct answer.

1) $(2, 4) \in R$ 2) $(3, 8) \in R$ ✓**3) $(6, 8) \in R$** 4) $(8, 7) \in R$

3. Let $f : R \to R$ be defined as $f(x) = x^4$. Choose the correct answer. [4]

1) f is one-one onto 2) f is many-one onto
3) f is one-one but not onto ✓**4) f is neither one-one nor onto.**

4. Let $f : R \to R$ be defined as $f(x) = 3x$. Choose the correct answer. [1]

✓**1) f is one-one onto** 2) f is many-one onto
3) f is one-one but not onto 4) f is neither one-one nor onto.

5. Let A = {1, 2, 3}. Then number of relations containing (1, 2) and (1, 3) which are reflexive and symmetric but not transitive is [1]

✓**1) 1** 2) 2 3) 3 4) 4

6. Let A = {1, 2, 3}. Then number of equivalence relations containing (1, 2) is [2]

1) 1 ✓**2) 2** 3) 3 4) 4

7. Set A has 3 elements and the set B has 4 elements. Then the number of injective mappings that can be defined from A to B : [3]

1) 144 2) 12 ✓**3) 24** 4) 64

8. Let A = {3, 5}. Then the number of reflexive relations on set A is [2]

1) 2 ✓**2) 4** 3) 0 4) 8

9. Let $f : R \to R$ be the function defined by $f(x) = \frac{2x-1}{2}$ and $g : R \to R$ be [2]
another function defined by $g(x) = x+2$, then $(gof)\left(\frac{3}{2}\right)$ is

1) 2 ✓**2) 3** 3)7/2 4)5/2

10. If $f(x) = \frac{2x+3}{4}$ then $f^{-1}(x)$ is [4]

1) $\frac{4}{2x+3}$ 2) $\frac{2x-3}{4}$ 3) $2x+3$ ✓4) $\frac{4x-3}{2}$

[☞ Answers for MCQ are kept in **bold font**]

Mahesh

2) INVERSE TRIG. & FUNCTIONS

I. Select the correct option from the given choices.

1. If $\sin^{-1}x = y$, then [2]

1) $0 \le y \le \pi$ ✓2) $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$ 3) $0 < y < \pi$ 4) $-\frac{\pi}{2} < y < \frac{\pi}{2}$

2. $\tan^{-1}\sqrt{3} - \sec^{-1}(-2)$ is equal to [2]

1) π ✓2) $-\frac{\pi}{3}$ 3) $\frac{\pi}{3}$ 4) $\frac{2\pi}{3}$

3. $\cos^{-1}\left(\cos\frac{7\pi}{6}\right)$ is equal to [2]

1) $\frac{7\pi}{6}$ ✓2) $\frac{5\pi}{6}$ 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{6}$

4. $\sin\left(\frac{\pi}{3} - \sin^{-1}\left(-\frac{1}{2}\right)\right)$ is equal to [4]

1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{1}{4}$ ✓4) 1

5. $\tan^{-1}\sqrt{3} - \cot^{-1}(-\sqrt{3})$ is equal to [2]

1) π ✓2) $-\frac{\pi}{2}$ 3) 0 4) $2\sqrt{3}$

6. $\sin\left[\frac{\pi}{3} + \sin^{-1}\left(\frac{-1}{2}\right)\right]$ is equal to : [2]

1) 1 ✓2) $\frac{1}{2}$ 3) $\frac{1}{3}$ 4) $\frac{1}{4}$

7. The principal value of $\cos^{-1}\left(\frac{1}{2}\right) + \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$ is [1]

✓1) $\frac{\pi}{12}$ 2) π 3) $\frac{\pi}{3}$ 4) $\frac{\pi}{6}$

8. The principal value of $\tan^{-1}\left(\tan\frac{9\pi}{8}\right)$ is [1]

✓1) $\frac{\pi}{8}$ 2) $\frac{3\pi}{8}$ 3) $-\frac{\pi}{8}$ 4) $-\frac{3\pi}{8}$

9. The domain of the function $\cos^{-1}(2x-3)$ is [4]

1) $[-1,1]$ 2) $(1,2)$ 3) $(-1,1)$ ✓4) $[1,2]$

10. The value of $\sin^{-1}\left(\cos\frac{3\pi}{5}\right)$ is [3]

1) $\frac{\pi}{10}$ 2) $\frac{3\pi}{5}$ ✓3) $-\frac{\pi}{10}$ 4) $-\frac{3\pi}{5}$

11. The principal value of $\tan^{-1}(-1)$ is [2]

1) $\frac{\pi}{4}$ ✓2) $-\frac{\pi}{4}$ 3) $\frac{\pi}{2}$ 4) $\frac{\pi}{3}$

12. The principal value of $\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$ is [4]

1) $\frac{13\pi}{6}$ 2) $\frac{\pi}{2}$ 3) $\frac{\pi}{3}$ ✓4) $\frac{\pi}{6}$

Mahesh

3) MATRICES

I. Select the correct option from the given choices.

1. $A=[a_{ij}]_{m\times n}$ is a square matrix, if [3]

1) $m<n$ 2) $m>n$ ✓**3) $m=n$** 4) $m=n^2$

2. Which of the given values of x and y make the following pair of matrices equal $\begin{bmatrix}3x+2 & 5\\ y+1 & 2-3x\end{bmatrix}=\begin{bmatrix}0 & y-2\\ 8 & 4\end{bmatrix}$ [3]

1) $x=\frac{-1}{3}, y=7$ 2) $y=7, x=\frac{2}{3}$ ✓**3) $y=7, x=\frac{-2}{3}$** 4) $x=\frac{-1}{3}, y=\frac{-2}{3}$

3. The number of all possible matrices of order 3 × 3 with each entry 0 or 1 is: [4]

1) 27 2) 18 3) 81 ✓**4) 512**

4. Assume X, Z are matrices of order 2 × n, 2 × p respectively. [2]
If n=p, then the order of the matrix 7X–5Z is

1) p×2 ✓**2) 2×n** 3) n×3 4) p×n

5. If A, B are symmetric matrices of same order, then AB – BA is a [1]

✓**1) Skew symmetric matrix** 2) Symmetric matrix

3) Zero matrix 4) Identity matrix

6. If $A=\begin{bmatrix}\cos\alpha & -\sin\alpha\\ \sin\alpha & \cos\alpha\end{bmatrix}$ and $A+A'=I$, then the value of α is **[BMP]** [2]

1) $\frac{\pi}{6}$ ✓**2) $\frac{\pi}{3}$** 3) π 4) $\frac{3\pi}{2}$

7. Matrices A and B will be inverse of each other only if [4]

1) AB = BA 2) AB = BA = O 3) AB = O, BA = I ✓**4) AB = BA = I**

8. If $A=\begin{bmatrix}\alpha & \beta\\ -\gamma & \alpha\end{bmatrix}$ is such that $A^2=I$, then [3]

1) $1+\alpha^2+\beta\gamma=0$ 2) $1-\alpha^2+\beta\gamma=0$ ✓**3) $1-\alpha^2-\beta\gamma=0$** 4) $1+\alpha^2-\beta\gamma=0$

9. If the matrix A is both symmetric and skew symmetric, then [2]

1) A is a diagonal matrix ✓**2) A is a zero matrix**

3) A is a square matrix 4) None of these

10. If A is square matrix such that $A^2=A$, then $(I+A)^3-7A$ is equal to [3]

1) A 2) I – A ✓**3) I** 4) 3A

Mahesh

4. DETERMINANTS

1. If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to [2]
1) 6 ✓2) ± 6 3) – 6 4) 0

2. If A is 3×3 matrix and det (3A) = k (detA), then k = [4]
1) 9 2) 6 3) 1 ✓4) 27

3. Value of k, for which $\begin{bmatrix} k & 2 \\ 8 & k \end{bmatrix}$ is singular [3]
1) 4 2) –4 ✓3) ± 4 4) 0

4. The area of a triangle with vertices (–3, 0), (0, 3) and (0, k) is 9 sq. units [1]
the value of k will be
✓1) 9 2) 3 3) –9 4) 6

5. If A is square matrix of order 3 and |A| = –4, then |adj A| is equal to [4]
1) –4 2) 4 3) –16 ✓4) 16

6. If area of triangle is 35 sq units with vertices (2, – 6), (5, 4) and (k, 4). [4]
Then k is
1) 12 2) –2 3) –12, –2 ✓ 4) 12, –2

7. If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ and A_{ij} is Cofactors of a_{ij} then value of Δ is given by [4]
1) $a_{11}A_{31} + a_{12}A_{32} + a_{13}A_{33}$ 2) $a_{11}A_{11} + a_{12}A_{21} + a_{13}A_{31}$
3) $a_{21}A_{11} + a_{22}A_{12} + a_{23}A_{13}$ ✓4) $a_{11}A_{11} + a_{21}A_{21} + a_{31}A_{31}$

8. Let A be a nonsingular square matrix of order 3 × 3. Then | adj A| is equal to [2]
1) |A| ✓2) $|A|^2$ 3) $|A|^3$ 4) 3|A|

9. If A is an invertible matrix of order 2, then det (A^{-1}) is equal to [2]
1) det (A) ✓2) $\frac{1}{\det(A)}$ 3) 1 4) 0

10. If x, y, z are nonzero real numbers, then the inverse of matrix $A = \begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$ is [1]

✓1) $\begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$ 2) $xyz\begin{bmatrix} x^{-1} & 0 & 0 \\ 0 & y^{-1} & 0 \\ 0 & 0 & z^{-1} \end{bmatrix}$ 3) $\frac{1}{xyz}\begin{bmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{bmatrix}$ 4) $\frac{1}{xyz}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

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5) CONTINUITY & DIFFERENTIABILITY

1. Which of the following statement is true? [4]
1) Every polynomial function is continuous 2) The function $f(x) = 5x + 3$ is continuous at $x = 0$
3) The function $f(x) = |x|$ is continuous at $x = 0$ ✓**4) All of the options are correct**

2. If $f(x) = \begin{cases} 3ax - 2b, & x > 1 \\ ax + b + 1, & x < 1 \end{cases}$ and $\lim_{x \to 1} f(x)$ exists. Then the relation between a and b is [2]
1) $3a - 2b = 1$ ✓**2) $2a - 3b = 1$** 3) $2a + 3b = 1$ 4) $2a + 3b = -1$

3. If a function f(x) defined on [a,b] is discontinuous at $x = \alpha \in [a,b]$. Then [2]
1) $\lim_{x \to \alpha^-} f(x) = \lim_{x \to \alpha^+} f(x) = f(\alpha)$ ✓**2) $\lim_{x \to \alpha} f(x) \neq f(\alpha)$**
3) $\lim_{x \to a^-} f(x) = f(a)$ 4) $\lim_{x \to b^+} f(x) = f(b)$

4. In the interval [0, 3]. The function $f(x) = |x-1| + |x-2|$ is [4]
1) discontinuous 2) differentiable
3) continuous but not differentiable at $x = 2$ only
✓**4) continuous but not differentiable at $x = 1$ and $x = 2$.**

5. If $y = \sqrt{x + \sqrt{x + \sqrt{x + \ldots\ldots\infty}}}$. Then $\frac{dy}{dx}$ is equal to [4]
1) $\frac{1}{y}$ 2) $\frac{1}{x}$ 3) $\frac{1}{2x-1}$ ✓ 4) $\frac{1}{2y-1}$

6. The set of all points where the function $f(x) = 2x|x|$ is differentiable is [1]
✓**1) $(-\infty, \infty)$** 2) $(-\infty, 0) \cup (0, \infty)$ 3) $(0, \infty)$ 4) $(-\infty, 0)$

7. If $y = \cos^{-1}(\cos x)$ then find $\frac{dy}{dx}$ at $x = \frac{5\pi}{4}$ [2]
1) 1 ✓**2) –1** 3) 0 4) $-\frac{1}{\sqrt{2}}$

8. If $f(x) = x^4 - x^3 + 7x^2 + 14$, then what is the value of $f^1(5)$? [4]
1) 594 2) 549 3) 954 ✓**4) 495**

9. $\frac{d}{dx}\left(e^{\log_e \sqrt{1+\tan^2 x}}\right)$ when $x \in Q_1$ **[BMP]** [3]
1) $\sec^2(x)\tan x$ 2) $\sec x \tan^2(x)$ ✓**3) secxtanx** 4) $\tan^2(x)$

10. If $y = \log(\cosh x)$ then $\frac{d^2y}{dx^2} =$ [1]
✓**1) $\text{sech}^2 x$** 2) $-\text{sech}^2 x$ 3) $\sinh x$ 4) $-\sinh x$

11. [x] represents the greatest integer function of x. At $x = -1$ $\frac{d}{dx}(\sin\pi[x]) =$ [1]
✓**1) 0** 2) 2 3) –2 4) 1/2

12. If $y = \log_y^x$ then $\frac{dy}{dx}$ is equal to [3]
1) $\frac{1}{x \log y}$ 2) $\frac{\log y}{x(1+\log y)}$ ✓**3) $\frac{1}{x(1+\log y)}$** 4) $\frac{1}{1+\log y}$

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6) APPLICATION OF DERIVATIVES

1. The rate of change of the area of a circle with respect to its radius r at r = 6 cm is [2]
   1) 10π ✓2) 12π 3) 8π 4) 11π

2. The total revenue in Rupees received from the sale of x units of a product is given by $R(x) = 3x^2 + 36x + 5$. The marginal revenue, when x = 15 is [4]
   1) 116 2) 96 3) 90 ✓4) 126

3. Which of the following functions are decreasing on $\left(0, \frac{\pi}{2}\right)$? [1]
   ✓1) cos x 2) cos 2x 3) cos 3x 4) tan x

4. In which interval $y = x^2 e^{-x}$ is increases [4]
   1) $(-\infty, \infty)$ 2) $(-2, 0)$ 3) $(2, \infty)$ ✓4) (0, 2)

5. On the curve $x^2 = 2y$ which is nearest to the point (0, 5) is [1]
   ✓1) $(2\sqrt{2}, 4)$ 2) $(2\sqrt{2}, 0)$ 3) (0, 0) 4) (2, 2)

6. For all real values of x, the minimum value of $\frac{1 - x + x^2}{1 + x + x^2}$ is [4]
   1) 0 2) 1 3) 3 ✓4) 1/3

7. A cylindrical tank of radius 10 m is being filled with wheat at the rate of 314 cubic metre per hour. Then the depth of the wheat is increasing at the rate of [1]
   ✓1) 1 m/h 2) 0.1 m/h 3) 1.1 m/h 4) 0.5 m/h

8. The function $f(x) = x^3 + 3x$ is increasing in interval [3]
   1) $(-\infty, 0)$ 2) $(0, \infty)$ ✓3) R 4) (0,1)

9. The interval in which the function $f(x) = 2x^3 + 9x^2 + 12x - 1$ is decreasing [2]
   1) $(-1, \infty)$ ✓2) $(-2, -1)$ 3) $(-\infty, -2)$ 4) $(-1, 1)$

10. At which point the function $f(x) = |x - 3|$ attains minimum value [3]
    1) $x = 1$ 2) $x < 3$ ✓3) $x = 3$ 4) $x > 3$

11. Minimum value of the function $f(x) = |x - 2| + |x - 5|$ is [3]
    1) 1 2) 2 ✓3) 3 4) 4

12. The maximum value of $\frac{\log x}{x}$ is $0 < x < \infty$ is [4]
    1) ∞ 2) e 3) 1 ✓4) e^{-1}

13. The minimum value of $(x - \alpha)(x - \beta)$ is [4]
    1) 0 2) $\alpha\beta$ 3) $\frac{1}{4}(\alpha - \beta)^2$ ✓4) $\frac{-1}{4}(\alpha - \beta)^2$

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7) INDEFINITE INTEGRALS & DEFINITE INTEGRALS

1) INDEFINITE INTEGRALS

1. The anti derivative of $e^{2\log\cot x}$ [3]

1) $\cot x - 1$ 2) $\tan x - \cot x$ ✓3) $-\cot x - x$ 4) $-1 - \cot x$

2. $\int \frac{10x^9 + 10^x \log_e 10}{x^{10} + 10^x} dx =$ [4]

1) $10^x - x^{10} + C$ 2) $10^x + x^{10} + C$ 3) $(10^x - x^{10})^{-1} + C$ ✓4) $\log(10^x + x^{10}) + C$

3. $\int \frac{dx}{x^2 + 2x + 2} =$ [2]

1) $x \tan^{-1}(x+1) + C$ ✓2) $\tan^{-1}(x+1) + C$ 3) $(x+1)\tan^{-1}x + C$ 4) $\tan^{-1}x + C$

4. $\int \frac{dx}{x(x^2+1)} =$ [1]

✓1) $\log|x| - \frac{1}{2}\log|x^2+1| + C$ 2) $\log|x| + \frac{1}{2}\log|x^2+1| + C$

3) $-\log|x| + \frac{1}{2}\log|x^2+1| + C$ 4) $\frac{1}{2}\log|x| + \log|x^2+1| + C$

5. $\int e^x \sec x (1 + \tan x) dx =$ [2]

1) $e^x \cos x + C$ ✓2) $e^x \sec x + C$ 3) $e^x \sin x + C$ 4) $e^x \tan x + C$

6. $\int e^x \left(\frac{1 + x\log x}{x}\right) dx =$ [2]

1) $xe^{\log x} + C$ ✓2) $e^x \log x + C$ 3) $e^x \log x^2 + C$ 4) None

2) DEFINITE INTEGRALS

7. $\int_1^{\sqrt{3}} \frac{dx}{1+x^2} =$ [4]

1) $\frac{\pi}{3}$ 2) $\frac{2\pi}{3}$ 3) $\frac{\pi}{6}$ ✓4) $\frac{\pi}{12}$

8. $\int_0^{\frac{2}{3}} \frac{dx}{4+9x^2} =$ [3]

1) $\frac{\pi}{6}$ 2) $\frac{\pi}{12}$ ✓3) $\frac{\pi}{24}$ 4) $\frac{\pi}{4}$

9. $\int_0^{\pi/2} \frac{3\sin x + 5\cos x}{\sin x + \cos x} dx =$ [1]

✓1) 2π 2) π 3) 4π 4) 8π

10. $\int_0^4 |2 - x| dx =$ [2]

1) 12 ✓2) 4 3) 8 4) 2

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8) APPLICATION OF INTEGRALS

1. Area of the region bounded by the curve $y^2 = 4x$, y-axis and the line $y = 3$ is [2]
   1) 2 ✓2) $\frac{9}{4}$ 3) $\frac{9}{3}$ 4) $\frac{9}{2}$

2. Area bounded by the curve $y = x^3$, the x-axis and the ordinates $x = -2$ and $x = 1$ is [4]
   1) -9 2) $\frac{-15}{4}$ 3) $\frac{15}{4}$ ✓4) $\frac{17}{4}$

3. Area under the curve $y = \sqrt{a^2 - x^2}$ included between the lines $x = 0$ and $x = a$ is [1]
   ✓1) $\frac{\pi a^2}{2}$ 2) $\frac{\pi a^2}{4}$ 3) $\frac{\pi a}{2}$ 4) $\frac{\pi a}{4}$

4. The area bounded by $y = \sin 2x$ the x - axis and the lines $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$ is [1]
   ✓1) 1 sq units 2) 2 sq. units 3) 4 sq. units 4) 3/ 2 sq.units

5. The area bounded by the curve $y = x^2 - 4$, and the lines $y = 0$ and $y = 5$ is [2]
   1) $\frac{38}{3}$ ✓2) $\frac{76}{3}$ 3) $\frac{16}{3}$ 4) $\frac{8}{3}$

6. The area of the region bounded by parabola $y^2 = 8x$ and latus rectum is [3] [BMP]
   1) $\frac{4}{3}$ 2) $\frac{16}{3}$ ✓3) $\frac{32}{3}$ 4) $\frac{8}{3}$

7. The area bounded by the curve $y = 2x - x^2$ and the line $y = -x$ is [3]
   1) 7/2 2) 7 ✓3) 9/2 4) 9

8. The area enclosed between the graph of $y = x^3$ and the lines $x = 0, y = 1, y = 8$ is [3]
   1) 7 2) 14 ✓3) 45/4 4) 54/4

9. The area of the region bounded by the curve $y^2 = x$, the Y - axis and between $y = 2$ and $y = 4$ [3]
   1) $\frac{52}{3}$ 2) $\frac{54}{3}$ ✓3) $\frac{56}{3}$ 4) $\frac{58}{3}$

10. Area of the region bounded by the curve $y = \cos x$ between $x = 0$ and $x = \pi$ and the X-axis is [2]
    1) 1 ✓2) 2 3) 3 4) 4

11. The area bounded by the curve $y = x^3$, X - axis and two ordinates $x = 1$ and $x = 2$ is
    1) $\frac{15}{2}$ ✓2) $\frac{15}{4}$ 3) $\frac{17}{2}$ 4) $\frac{17}{4}$ [2]

12. The area bounded by the curves $y^2 = 4x$ and $y = x$ is equal to [2]
    1) $\frac{1}{3}$ ✓2) $\frac{8}{3}$ 3) $\frac{35}{6}$ 4) $\frac{7}{3}$

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9) DIFFERENTIAL EQUATIONS

1. The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3+\left(\frac{dy}{dx}\right)^2+\sin\left(\frac{dy}{dx}\right)+1=0$ is [4]
1) 3 2) 2 3) 1 ✓4) not defined

2. The order of the differential equation $2x^2\frac{d^2y}{dx^2}-3\frac{dy}{dx}+y=0$ is [1]
✓1) 2 2) 1 3) 0 4) not defined

3. The general solution of the differential equation $dy/dx=e^{x+y}$ is [1]
✓1) $e^x+e^{-y}=C$ 2) $e^x+e^y=C$ 3) $e^{-x}+e^y=C$ 4) $e^{-x}+e^{-x}=C$

4. Which of the following is a homogeneous differential equation? [4]
1) $(4x+6y+5)\,dy-(3y+2x+4)\,dx=0$ 2) $(xy)\,dx-(x^3+y^3)\,dy=0$
3) $(x^3+2y^2)\,dx+2xy\,dy=0$ ✓4) $y^2\,dx+(x^2-xy-y^2)\,dy=0$

5. The Integrating Factor of the differential equation $x\frac{dy}{dx}-y=2x^2$ is [3]
1) e^{-x} 2) e^{-y} ✓3) $1/x$ 4) x

6. The Integrating Factor of the D.E $(1-y^2)\frac{dx}{dy}+yx=ay,\ (-1<y<1)$ is [4]
1) $\frac{1}{y^2-1}$ 2) $\frac{1}{\sqrt{y^2-1}}$ 3) $\frac{1}{1-y^2}$ ✓4) $\frac{1}{\sqrt{1-y^2}}$

7. The general solution of the differential equation $\frac{ydx-xdy}{y}=0$ is [3]
1) $xy=C$ 2) $x=Cy^2$ ✓3) $y=Cx$ 4) $y=Cx^2$

8. The general solution of the differential equation $e^x\,dy+(y\,e^x+2x)\,dx=0$ is [3]
1) $x\,e^y+x^2=C$ 2) $x\,e^y+y^2=C$ ✓3) $y\,e^x+x^2=C$ 4) $y\,e^y+x^2=C$

9. General solution of the differential equation $\log\left(\frac{dy}{dx}\right)=2x+y$ is [3]
1) $e^{-y}=\frac{1}{2}e^{2x}+C$ 2) $\frac{1}{e^y}+\frac{1}{2}e^{2x}=C$ ✓3) $-e^{-y}=\frac{1}{2}e^{2x}+C$ 4) $e^y=\frac{1}{2}e^{2x}+C$

10. General solution of differential equation $\frac{dy}{dx}=\frac{y}{x}$ is [2]
1) $\log y=Cx$ ✓2) $y=Cx$ 3) $xy=C$ 4) $y=C\log x$

11. The degree of the differential equation $\left(1+\frac{dy}{dx}\right)^3=\left(\frac{dy}{dx}\right)^2$ is [3]
1) 1 2) 2 ✓3) 3 4) 44

12. The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3+\left(\frac{dy}{dx}\right)^2+\sin\left(\frac{dy}{dx}\right)+1=0$ is [4]
1) 3 2) 2 3) 1 ✓4) not defined

13. Order of D.E correspoinding to family of curves $y=Ae^{2x}+Be^{-2x}$ is [1]
✓1) 2 2) 1 3) 3 4) 4

14. The general solution of D.E is $dy/dx=e^{x-y}$ is [1]
✓1) $e^y=e^x+C$ 2) $e^x+e^y=C$ 3) $e^{x+y}=C$ 4) $e^{x-y}=C$

15. The D.E for which $ax+by=1$ is general solution (a, b are arbitrary constants) is [3]
1) $\frac{dy}{dx}=x+C$ 2) $y\frac{d^2y}{dx^2}+x=1$ ✓3) $\frac{d^2y}{dx^2}=0$ 4) $\frac{d^3y}{dx^3}=0$

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10) VECTOR ALGEBRA

1. Area of a rectangle having vertices A, B, C and D with position vectors $-\hat{i}+\frac{1}{2}\hat{j}+4\hat{k}, \hat{i}+\frac{1}{2}\hat{j}+4\hat{k}$ and $-\hat{i}-\frac{1}{2}\hat{j}+4\hat{k}$, respectively is [3]
1) 1/2 2) 1 ✓3)2 4) 4

2. If θ is the angle between two vectors $\vec{a}$ and $\vec{b}$, then $\vec{a}.\vec{b} \geq 0$ only when [2]
1) $0<\theta<\frac{\pi}{2}$ ✓2) $0 \leq \theta \leq \frac{\pi}{2}$ 3) $0<\theta<\pi$ 4) $0 \leq \theta \leq \pi$

3. Let $\vec{a}$ and $\vec{b}$ be two unit vectors and θ is the angle between them. [4]
Then $\vec{a}+\vec{b}$ is a unit vector if
1) $\theta=\frac{\pi}{4}$ 2) $\theta=\frac{\pi}{3}$ 3) $\theta=\frac{\pi}{2}$ ✓4) $\theta=\frac{2\pi}{3}$

4. The value of $\hat{i}.(\hat{j}\times\hat{k})+\hat{j}.(\hat{i}\times\hat{k})+\hat{k}.(\hat{i}\times\hat{j})$ is [3]
1) 0 2) −1 ✓3)1 4)3

5. If θ is the angle between any two vectors $\vec{a}$ and $\vec{b}$, then $|\vec{a}.\vec{b}|=|\vec{a}\times\vec{b}|$ [2]
when θ is equal to
1) 0 ✓2) $\pi/4$ 3) $\pi/2$ 4)π

6. The value of the dot product of $\vec{a}-\vec{b}$ and $\vec{a}+\vec{b}$ is [1]
✓1)a^2-b^2 2) $(\vec{a}\times\vec{b})$ 3)$\vec{a}\times\vec{b}$ 4)$\vec{b}\times\vec{a}$

7. The position vector of the point $(1,2,0)$ is [3]
1) $\vec{i}+\vec{j}+\vec{k}$ 2) $\vec{i}+2\vec{j}+\vec{k}$ ✓3) $\vec{i}+2\vec{j}$ 4) $2\vec{j}+\vec{k}$

8. If $|(\vec{a}\times\vec{b})|=4$ and $|\vec{a}\cdot\vec{b}|=2$ then $|\vec{a}|^2|\vec{b}|^2$ is equal to [3]
1) 4 2) 2 ✓3)20 4)2

9. The points with position vectors $10\vec{i}+3\vec{j}$, $12\vec{i}-5\vec{j}$ and $a\vec{i}+11\vec{j}$ are collinear, if a is [4]
1) 2 2)−8 3)4 ✓4)8

10. The vector $\cos\alpha\cos\beta\ \vec{i}+\cos\alpha\sin\beta\ \vec{j}+\sin\alpha\ \vec{k}$ is [2]
1) null vector ✓2)unit vector 3)constant vector 4)vector with magnitude >1

11. If $\vec{a},\vec{b},\vec{c}$ are mutually perpendicular unit vectors, then the value of $|\vec{a}+\vec{b}+\vec{c}|$ is [3]
1)1 2) $\sqrt{2}$ ✓3) $\sqrt{3}$ 4)2

12. If $\vec{a}+\vec{b}+\vec{c}=\vec{0}, |\vec{a}|=3, |\vec{b}|=5, |\vec{c}|=7$ then the angle between $\vec{a}$ and $\vec{b}$ is [4]
1) $\frac{\pi}{6}$ 2) $\frac{2\pi}{3}$ 3) $\frac{5\pi}{3}$ ✓4) $\frac{\pi}{3}$

13. If $\vec{a}$, and $\vec{b}$ are two unit vectors inclined at an angle θ then the value of $|\vec{a}-\vec{b}|$ is [1]
✓1) $2\sin\frac{\theta}{2}$ 2) $2\sin\theta$ 3) $2\cos\frac{\theta}{2}$ 4) $2\cos\theta$

14. If $|\vec{a}|=3$ and $-1 \leq k \leq 2$ then $|k\vec{a}|$ lies in the internal [1]
✓1) $[0,6]$ 2) $[-3,6]$ 3) $[3,6]$ 4) $[1,2]$

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11) LINEAR PROGRAMMING

1. Region represented by $x \geq 0, y \geq 0$ is [1]
✓**1) First quadrant** 2) Second quadrant 3) Third quadrant 4) Fourth quadrant

2. If the objective function $Z = ax + by$ has both a maximum and a minimum value on the region R then R is [1]
✓**1) Bounded** 2) Unbounded 3) Concave polygon 4) Infeasible

3. The linear inequalities or equations or restrictions on the variables of a linear programming problem are called [1]
✓**1) constriants** 2) Decision variables 3) objective function 4) Linear relations

4. The optimal value of the objective function is attained at the points [3]
1) On X-axis 2) On Y-axis **[BMP]**
✓**3) Which are at the corner points of the feasible region**
4) Which are at the points of intersection of the inequation with Y-axis

5. In a linear programming problem the objective function and constraints must be [2]
1) Non-linear ✓**2) Linear** 3) Exponential 4) Logarithmic

6. The maximum value of $Z = 3x + 4y$ subject to constraints $x + y \leq 4$, $x \geq 0, y \geq 0$ is [3]
1) 12 2) 14 ✓**3) 16** 4) 10

7. Maximize $Z = 3x + 5y$, subject to constriants $x + 4y \leq 24$, $3x + y \leq 21$, $x + y \leq 9, x \geq 0, y \geq 0$. [3]
1) 20 at (1, 0) 2) 30 at (0, 6) ✓**3) 37 at (4, 5)** 4) 33 at (6, 3)

8. The point which does not lie in the half plane $2x + 3y - 12 < 0$ is [4]
1) (2, 1) 2) (1, 2) 3) (–2, 3) ✓**4) (2, 3)**

9. Which of the following is not a component of linear programming problem? [4]
1) Objective function 2) Constriant
3) Decision variable ✓**4) Differential equation**

10. The position of points O(0, 0) and P(2, –3) in the region of graph of inequation $2x - 3y < 5$ will be [1]
✓**1) O inside and P outside** 2) O and P both inside
3) O and P both outside 4) O outside and P inside

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12) PROBABILITY

1. If P(A)=1/2, P(B)=0, then P(A|B) is [3]
1) 0 2)1/2 ✓3) not exist 4) 1

2. If A and B are events such that P(A|B) = P(B|A), then [4]
1) A ⊂ B but A ≠ B 2) A = B 3) A ∩ B = ϕ ✓4) P(A) = P(B)

3. The probability of obtaining an even prime number on each die, when a pair of dice is rolled is [4]
1)0 2) 1/3 3)1/12 ✓4)1/36

4. If A and B are two events such that P(A) ≠ 0 and P(B | A) = 1, then **[BMP]** [1]
✓1) A ⊂ B 2) B ⊂ A 3) B = ϕ 4) A = ϕ

5. If P(A|B) > P(A), then which of the following is correct : [3]
1) P(B|A) < P(B) 2) P(A ∩ B) < P(A) . P(B) ✓3) P(B|A) > P(B) 4) P(B|A) = P(B)

6. If A and B are any two events such that P(A) + P(B) – P(A and B) = P(A), then [2]
1) P(B|A) = 1 ✓2) P(A|B) = 1 3) P(B|A) = 0 4) P(A|B) = 0

7. A bag contains 7 red and 3 white balls. Three balls are drawn one after other without replacement. Then the probability that the first two are red and third one is white is [1]
✓1) 7/40 2) 33/40 3)23/40 4) 17/40

8. A book consists of 20 pages. If two pages are drawn (opened) at random, then the probability that both numbers are prime numbers is [4]
1)17/95 2)16/95 3)2/15 ✓4)14/95

9. A fair coin is tossed 3 times, then the probability of getting one head and two tails is [3]
1)1/8 2)1/4 ✓3)3/8 4)1/2

10. A person appears for an interview for two posts A and B. The selection of the posts are independent. If $P(A)=\frac{1}{5}, P(B)=\frac{1}{8}$ then $P(A \cup B)$ is [2]
1)7/10 ✓2)3/10 3)9/10 4)1/10

11. Three events A, B, C are mutually exclussive and exhaustive and P(A) = 0.4, then P(B) + P(C) = [3]
1) 0.4 2) 0.5 ✓3) 0.6 4) 0

12. If P(A ∪B) = 0.65 and P(A ∩ B) = 0.15, then $P(\bar{A})+P(\bar{B})$ = [3]
1) 0.8 2) 0.6 ✓3) 1.2 4) 1.4

13. When two dice are rolled, the probability of getting unequal numbers on the faces is [3]
1) 1/6 2) 35/36 ✓3) 5/6 4) 1/3

14. A fair coin whose faces are marked with 1 and 2 is thrown for four times. Then the probability of throwing a total of atleast 5 is [3]
1) 1/16 2) 5/16 ✓3) 15/16 4) 3/16

15. A fair die is rolled. Consider the events A = {1, 3, 5} and B = {2, 3}, then P(A|B) is [1]
✓1) $\frac{1}{2}$ 2) $\frac{1}{3}$ 3) $\frac{2}{3}$ 4) $\frac{5}{6}$

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LILLIPUT 1 MARK BIT QUESTIONS(Q7 to Q12)

INVERSE TRIG. FUNCTIONS

I. Find the principal values of 1) $\sin^{-1}\left(-\frac{1}{2}\right)$ 2) $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$ 3) $\cos^{-1}\left(-\frac{1}{2}\right)$ 4) $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$

5) $\tan^{-1}(-1)$ 6) $\tan^{-1}(-\sqrt{3})$ 7) $\text{cosec}^{-1}(2)$ 8) $\text{cosec}^{-1}\left(-\sqrt{2}\right)$ 9) $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right)$ 10) $\cot^{-1}(\sqrt{3})$

Sol:

1. We know that $\sin^{-1}(-x) = -\sin^{-1}x$

$\therefore \sin^{-1}\left(\frac{-1}{2}\right) = -\sin^{-1}\left(\frac{1}{2}\right) = -\sin^{-1}\left[\sin\left(\frac{\pi}{6}\right)\right] = -\frac{\pi}{6}$ Here $-\frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$

2. $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \cos^{-1}\left[\cos\left(\frac{\pi}{6}\right)\right] = \frac{\pi}{6}$ Here $\frac{\pi}{6} \in [0, \pi]$

3. We know that $\cos^{-1}(-x) = \pi - \cos^{-1}x$ **[BMP]**

$\cos^{-1}\left(\frac{-1}{2}\right) = \pi - \cos^{-1}\left(\frac{1}{2}\right) = \pi - \cos^{-1}\left[\cos\left(\frac{\pi}{3}\right)\right] = \pi - \frac{\pi}{3} = \frac{2\pi}{3}$. Here $-\frac{\pi}{3} \in [0, \pi]$

4. $\cos^{-1}\left(\frac{-1}{\sqrt{2}}\right) = \pi - \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = \pi - \cos^{-1}\left[\cos\left(\frac{\pi}{4}\right)\right] = \pi - \frac{\pi}{4} = \frac{3\pi}{4}$ Here $\frac{3\pi}{4} \in [0, \pi]$

5. We know that $\tan^{-1}(-x) = -\tan^{-1}x$

$\tan^{-1}(-1) = -\tan^{-1}1 = -\tan^{-1}\left[\tan\left(\frac{\pi}{4}\right)\right] = -\frac{\pi}{4}$ Here $-\frac{\pi}{4} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

6. $\tan^{-1}\left(-\sqrt{3}\right) = -\tan^{-1}(\sqrt{3}) = -\tan^{-1}\left(\tan\frac{\pi}{3}\right) = -\frac{\pi}{3}$ Here $-\frac{\pi}{3} \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

7. $\text{cosec}^{-1}(2) = \text{cosec}^{-1}\left[\text{cosec}\left(\frac{\pi}{6}\right)\right] = \frac{\pi}{6}$ Here $\frac{\pi}{6} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

8. We know that $\text{cosec}^{-1}(-x) = -\text{cosec}^{-1}x$

$\text{cosec}^{-1}(-\sqrt{2}) = -\text{cosec}^{-1}(\sqrt{2}) = -\text{cosec}^{-1}\left[\text{cosec}\left(\frac{\pi}{4}\right)\right] = -\frac{\pi}{4}$ Here $-\frac{\pi}{4} \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] - \{0\}$

9. $\sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \sec^{-1}\left(\sec\frac{\pi}{6}\right) = \frac{\pi}{6}$ Here $\frac{\pi}{6} \in [0, \pi] - \{\frac{\pi}{2}\}$

10. $\cot^{-1}(\sqrt{3}) = \cot^{-1}\left[\cot\left(\frac{\pi}{6}\right)\right] = \frac{\pi}{6}$ Here $\frac{\pi}{6} \in (0, \pi)$

Mahesh

*11. Find the value of $\cos^{-1}\left(\frac{1}{2}\right)+2\sin^{-1}\left(\frac{1}{2}\right)$

Sol: $\cos^{-1}\left[\cos\frac{\pi}{3}\right]+2\sin^{-1}\left[\sin\frac{\pi}{6}\right]=\frac{\pi}{3}+2\left(\frac{\pi}{6}\right)=\frac{\pi}{3}+\frac{\pi}{3}=\frac{2\pi}{3}$. Here $\frac{2\pi}{3}\in[0,\pi]$

12. Find the value of $\tan^{-1}(1)+\cos^{-1}\left(-\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)$

Sol: We know that $\cos^{-1}(-x)=\pi-\cos^{-1}x$ and $\sin^{-1}(-x)=-\sin^{-1}x$

$\therefore \tan^{-1}(1)+\cos^{-1}\left(-\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)=\tan^{-1}(1)+\left[\pi-\cos^{-1}\left(\frac{1}{2}\right)\right]-\sin^{-1}\left(\frac{1}{2}\right)$

$=\frac{\pi}{4}+\left(\pi-\frac{\pi}{3}\right)-\frac{\pi}{6}=45^\circ+180^\circ-60^\circ-30^\circ=135^\circ=\frac{3\pi}{4}$

MATRICES & DETERMINANTS

13. Define a Symmetric Matrix.

Sol: SYMMETRIC MATRIX: A square matrix A is said to be a Symmetric matrix if $A'=A$.

Ex: $A=\begin{bmatrix}1 & 2\\ 2 & 3\end{bmatrix}$

14. Define Skew Symmetric Matrix.

Sol: SKEW-SYMMETRIC MATRIX:

A square matrix A is said to be a skew-symmetric matrix if $A'=-A$. Ex: $A=\begin{bmatrix}0 & -1\\ 1 & 0\end{bmatrix}$

15. Define Inverse of a Matrix.

Sol: INVERSE OF A MATRIX: A square matrix A is said to be an invertible matrix if there exists a square matrix B such that AB=BA=I and the matrix B is called inverse of A.

- If A is a nonsingular matrix then $A^{-1}=\frac{1}{|A|}(\text{adj } A)$

VECTOR ALGEBRA

16. Compute the magnitude of the vector $\vec{c}=\frac{1}{\sqrt{3}}\hat{i}+\frac{1}{\sqrt{3}}\hat{j}-\frac{1}{\sqrt{3}}\hat{k}$

Sol: Given that $\vec{c}=\frac{1}{\sqrt{3}}\hat{i}+\frac{1}{\sqrt{3}}\hat{j}-\frac{1}{\sqrt{3}}\hat{k}$ $|\vec{c}|=\sqrt{\left(\frac{1}{\sqrt{3}}\right)^2+\left(\frac{1}{\sqrt{3}}\right)^2+\left(\frac{-1}{\sqrt{3}}\right)^2}=\sqrt{\frac{1}{3}+\frac{1}{3}+\frac{1}{3}}=\sqrt{\frac{3}{3}}=\sqrt{1}=1.$

Mahesh

17. Find the angle between the vectors $\hat{i}-2\hat{j}+3\hat{k}$ and $3\hat{i}-2\hat{j}+\hat{k}$.

Sol: Given that $\vec{a}=\hat{i}-2\hat{j}+3\hat{k}$ and $\vec{b}=3\hat{i}-2\hat{j}+\hat{k}$.

$\Rightarrow |\vec{a}|=\sqrt{1+4+9}=\sqrt{14}$ and $|\vec{b}|=\sqrt{9+4+1}=\sqrt{14}$

Also $\vec{a}.\vec{b}=(\hat{i}-2\hat{j}+3\hat{k}).(3\hat{i}-2\hat{j}+\hat{k})=1(3)+(-2)(-2)+3(1)=3+4+3=10$

Let θ be the angle between the vectors $\vec{a}$ and $\vec{b}$.

$$\therefore \cos\theta=\frac{\vec{a}.\vec{b}}{|\vec{a}||\vec{b}|}=\frac{10}{\sqrt{14}\sqrt{14}}=\frac{10}{14}=\frac{5}{7}\Rightarrow\theta=\cos^{-1}\left(\frac{5}{7}\right)$$

PROBABILITY

18. Define Conditional Probability.

A: **Conditional Probability:** The conditional probability of an event E, given the occurrence of the event F is given by $P(E/F)=\frac{P(E\cap F)}{P(F)}, P(F)\neq 0$

---

19. Let E and F be events with $P(E)=\frac{3}{5}$, $P(F)=\frac{3}{10}$ and $P(E\cap F)=\frac{1}{5}$. Are E and F independent?

Sol: Given $P(E\cap F)=\frac{1}{5}$ Now $P(E)P(F)=\frac{3}{5}\times\frac{3}{10}=\frac{9}{50}\neq\frac{1}{5}$

$P(E)P(F)\neq P(E\cap F)$ $\therefore$ E and F are not independent.

LINEAR PROGRAMMING

20. Define Objective function

Sol: **Objective function :** The Linear function $Z = ax + by$, where a, b are constants, which has to be maximized or minimized in the LPP is called a linear **objective function.**

---

21. Define Constraints

Sol: **Constraints:** The linear inequalities or equations or restrictions on the variables of a linear programming problem are called **constraints**.

---

22. Define Optimal Solution

Sol: **Optimal (feasible) solution :** The point in the feasible region that gives the optimal value (maximum or minimum) of the objective function is called an **optimal solution.**

Mahesh

DIFFERENTIAL EQUATIONS

23. Determine order and degree of $y'+5y=0$

Sol: Given D.E is $y'+5y=0$

Highest order derivative in the D.E is y' . Its order is one.

It is a polynomial equation in y'. Highest power of y' is 1. So degree of the D.E is one.

---

24. Determine order and degree of $\frac{d^2y}{dx^2}=\cos 3x+\sin 3x$

Sol: Highest order derivative in the given D.E is $\frac{d^2y}{dx^2}$. Its order is two.

It is a polynomial equation in $\frac{d^2y}{dx^2}$ and the power is 1. So degree of D.E is 1.

---

25. Determine order and degree of $\frac{d^4y}{dx^4}+\sin(y'')=0$

Sol: Given D.E is $\frac{d^4y}{dx^4}+\sin(y'')=0 \Rightarrow y''''+\sin(y'')=0$

Highest order derivative in the differential equation is y'''' . Its order is four.

But the D.E is not a polynomial equation in its derivatives. So its degree is not defined.

---

26. Determine order and degree of $\left(\frac{d^2y}{dx^2}\right)^2+\cos\left(\frac{dy}{dx}\right)=0$

Sol: Highest order derivative in the given D.E is $\frac{d^2y}{dx^2}$. Its order is 2.

Given differential equation is not a polynomial equation in its derivatives.

Degree of the D.E is not defined.

Mahesh

GUESS PAPERS

Q1 BOARD MODEL PAPER

Time : 3 Hours **SR. MATHS** Max.Marks : 100

SECTION - A

Ans-Page Index

I. Answer ALL the questions. Each question carries ONE mark. $12 \times 1 = 12$

1. Let R be the relation in the set {1, 2, 3, 4} given by R = {(1, 2), (2, 2), (1, 1), (4,4), (1, 3), (3, 3), (3, 2)}. Choose the correct answer. |☞P 131(1)|
   1) R is reflexive and symmetric but not transitive.
   2) R is reflexive and transitive but not symmetric.
   3) R is symmetric and transitive but not reflexive. 4)R is an equivalence relation.

2. **If $A = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$ and $A + A' = I$, then the value of α is** |☞P 133(6)|
   1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) π 4) $\frac{3\pi}{2}$

3. $\frac{d}{dx}\left(e^{\log_e \sqrt{1+\tan^2 x}}\right)$ **when** $x \in Q_1$ |☞P 135(9)|
   1) $\sec^2(x)\tan x$ 2) $\sec x \tan^2(x)$ 3) secxtanx 4) $\tan^2(x)$

4. **The area of the region bounded by parabola $y^2 = 8x$ and latus rectum is** |☞P 138(6)|
   1) $\frac{4}{3}$ 2) $\frac{16}{3}$ 3) $\frac{32}{3}$ 4) $\frac{8}{3}$

5. The optimal value of the objective function is attained at the points |☞P 141(4)|
   1) On X-axis 2) On Y-axis
   3) Which are at the corner points of the feasible region
   4) Which are at the points of intersection of the inequation with Y-axis

6. If A and B are two events such that $P(A) \neq 0$ and $P(B \mid A) = 1$, then |☞P 142(4)|
   1) $A \subset B$ 2) $B \subset A$ 3) $B = \phi$ 4) $A = \phi$

7. Find the principal value of $\cos^{-1}\left(\frac{1}{2}\right)$ |☞P 143(3)|

8. Find adjA for $A = \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix}$ |☞P 104(187)|

9. Find the rate of change of the area of a circle with respect to its radius r when r = 3 cm |☞P 119(253)|

10. Find $\int \text{cosecx}(\text{cosecx}+\cot x)dx$ |☞P 120(PQ)|

11. Determine order and degree of $\left(\frac{ds}{dt}\right)^4 + 3s\frac{d^2s}{dt^2} = 0$ |☞P 125(281)|

12. Compute the magnitude of the vector $\vec{a} = \hat{i} + \hat{j} + \hat{k}$ |☞P 105(190)|

Mahesh

SECTION-B

II. Answer ALL the questions. $10 \times 2 = 20$
Each question carries TWO marks.

13. Check the injectivity and surjectivity of the function $f: N \to N$ given by $f(x) = x^3$ [☞P 96(155)]

14. Find the values of x,y and z from $\begin{bmatrix} x+y & 2 \\ 5+z & xy \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 5 & 8 \end{bmatrix}$ [☞P 99(169)]

15. Find equation of line joining (1, 2) and (3, 6) using determinants. [☞P 68(90)]

16. Differentiate the function cos(sin x) with respect to x . [☞P 115(232)]

17. Show that the function f given by $f(x) = x^3 - 3x^2 + 4x, x \in R$ is increasing on R. [☞P 118(246)]

18. Integrate the function $\frac{(\log x)^2}{x}$ [☞P 121(262)]

19. Verify that the function $y = Ax$ is a solution of the corresponding differential equation to $xy' = y (x \neq 0)$ [☞P 126(284)]

20. Classify the following as scalar and vector quantities i)force ii) distance

21. If the points (2, –1, 3), (4, *a*, 1) and (3, 1, *b*) are collinear, then find the ratio between *a* and *b*? [☞P 110(213)]

22. A and B are two events such that $P(A) \neq 0$. Find P(B|A), if [☞P 112(222)]
(i)A is a subset of B (ii) $A \cap B = \phi$

SECTION-C

III. Answer ANY SEVEN the questions. $7 \times 4 = 28$
Each question carries FOUR mark.

23. Show that each of the relation R in the set $A = \{x \in Z : 0 \leq x \leq 12\}$, given by $R = \{(a, b): |a - b|$ is a multiple of 4$\}$ is an equivalence relation. Find the set of all elements related to 1. [☞P 93(4)]

24. Write the function $\tan^{-1}\frac{\sqrt{1+x^2}-1}{x}, x \neq 0$ in the simplest form [☞P 59(74)]

25. Express the matrix $\begin{bmatrix} 3 & 3 & -1 \\ -2 & -2 & 1 \\ -4 & -5 & 2 \end{bmatrix}$ as the sum of a symmetric and a skew symmetric matrix [☞P 62(78)]

26. Show that $\begin{vmatrix} y+z & x & x \\ y & z+x & y \\ z & z & x+y \end{vmatrix} = 4xyz$ [☞P 72(98)]

27. Find $\frac{dy}{dx}$ of $y = \sin^{-1}\left(\frac{1-x^2}{1+x^2}\right), 0 < x < 1$ [☞P 81(12

28. Find the integral of $\frac{3x-1}{(x-1)(x-2)(x-3)}$ [☞P 85(13

29. Sketch the graph of $y = |x+3|$ and evaluate $\int_{-6}^{0} |x+3| dx$. [☞P 94(1

Ans-Page Index

Naresh

30. Show that the points $A(2\hat{i}-\hat{j}+\hat{k})$, $B(\hat{i}-3\hat{j}-5\hat{k})$, $C(3\hat{i}-4\hat{j}-4\hat{k})$ are the vertices of a right angled triangle. [☞P 74(104)]

31. Solve the following linear programming problem graphically: [☞P 79(117)]
Minimise $Z = 200x + 500y$ ..... (1) subject to the constraints:
$x + 2y \geq 10$.... (2) $3x + 4y \leq 24$.... (3) $x \geq 0$, $y \geq 0$.... (4)

32. Given two independent events A and B such that $P(A) = 0.3$, $P(B) = 0.6$. [☞P 75(106)]
Find (i) P(A and B)(ii) P(A and not B)(iii) P(A or B)(iv) P(neither A nor B)

SECTION-D

IV. Answer ANY FIVE the questions. $5 \times 8 = 40$

33. Solve the system of equations using matrix method [☞P 17(2)]
$2x + 3y + 3z = 5$, $x - 2y + z = -4$, $3x - y - 2z = 3$

34. Sand is pouring from a pipe at the rate of 12 cm^3/s. The falling sand forms a cone on the ground in such a way that the height of the cone is always one-sixth of the radius of the base. How fast is the height of the sand cone increasing when the height is 4 cm? [☞P 30(25)]

35. Find the integral of $\frac{x+3}{x^2-2x-5}$ [☞P 41(45)]

36. Evaluate $\int_0^{\pi} \frac{x}{1+\sin x}dx$ [☞P 35(34)]

37. Find the equation of a curve passing through the origin given that the slope of the tangent to the curve at any point (x, y) is equal to the sum of the coordinates of the point. [☞P 47(55)]

38. If $\bar{a} = 2\bar{i} + 3\bar{j} + 4\bar{k}$, $\bar{b} = \bar{i} + \bar{j} - \bar{k}$, $\bar{c} = \bar{i} - \bar{j} + \bar{k}$, compute $\bar{a} \times (\bar{b} \times \bar{c})$ and verify that it is perpendicular to $\bar{a}$ [☞P 19(5)]

39. Find the shortest distance between the lines [☞P 22(11)]
$\vec{r} = (\hat{i} + 2\hat{j} + \hat{k}) + \lambda(\hat{i} - \hat{j} + \hat{k})$ and $\vec{r} = 2\hat{i} - \hat{j} - \hat{k} + \mu(2\hat{i} + \hat{j} + 2\hat{k})$

40. Suppose we have four boxes A,B,C, D containing coloured marbles as given below: [☞P 29(24)]

Box	Marble colour		
	Red	White	Black
A	1	6	3
B	6	2	2
C	8	1	1
D	0	6	4

One of the boxes has been selected at random and a single marble is drawn from it.

Manesh

Q 2

GUESS PAPERS

GUESS PAPER -1

SR. MATHS

Time : 3 Hours | Max.Marks : 100

Ans-Page Index

SECTION-A

I. **Answer ALL the questions.** $12 \times 1 = 12$

Each question carries ONE mark.

1)* Let $f : R \to R$ be defined as $f(x) = 3x$. Choose the correct answer. [☞P 131(4)]

1) f is one-one onto 2) f is many-one onto

3) f is one-one but not onto 4) f is neither one-one nor onto.

2) If A is square matrix such that $A^2 = A$, then $(I + A)^3 - 7A$ is equal to [☞P 133(10)]

1) A 2) I – A 3) I 4) 3A

3)* If $f(x)=x^4-x^3+7x^2+14$, then what is the value of $f^1(5)$? [☞P 135(8)]

1) 594 2) 549 3) 954 4) 495

4)* $\int \frac{dx}{x^2+2x+2} =$ [☞P 137(3)]

1) $x\tan^{-1}(x+1)+C$ 2) $\tan^{-1}(x+1)+C$ 3) $(x+1)\tan^{-1}x+C$ 4) $\tan^{-1}x+C$

5) If $|(\vec{a}\times\vec{b})|=4$ and $|\vec{a}\cdot\vec{b}|=2$ then $|\vec{a}|^2 |\vec{b}|^2$ is equal to [☞P 140(8)]

1) 4 2) 2 3) 20 4) 2

6) The probability of obtaining an even prime number on each die, when a pair of dice is rolled is [☞P 142(3)]

1) 0 2) 1 3) $\frac{1}{12}$ 4) $\frac{1}{36}$

7) Find the value of $\tan^{-1}(1)+\cos^{-1}\left(-\frac{1}{2}\right)+\sin^{-1}\left(-\frac{1}{2}\right)$ [☞P 144(12)]

8) Evaluate $\begin{vmatrix} x & x+1 \\ x-1 & x \end{vmatrix}$ [☞P 103(181)]

9) Find the intervals in which the function f given by $f(x) = x^2 - 4x + 6$ is strictly increasing [☞P 118(248)]

10) Find the area under the given curves and given lines: [☞P 90(142)]

$y = x^2$, $x = 1$, $x = 2$ and x-axis

11)* Find the general solution of the differential equation $\frac{dy}{dx}=\frac{1+y^2}{1+x^2}$ [☞P 127(28

12)* Define Objective function [☞P 145(20

☞ **Guess Paper 1** is unofficial BMP1 in which * Problems are modified

Mahesh

SECTION-B

II. Answer ALL the questions.

Each question carries TWO marks. $10 \times 2 = 20$

13) Check whether the relation R defined in the set {1, 2, 3, 4, 5, 6} as $R = \{(a, b): b = a+1\}$ is reflexive, symmetric or transitive. [☞P 54(62)]

14) Construct a 3×2 matrix whose elements are given by $a_{ij} = \frac{1}{2}|i - 3j|$ [☞P 100(PQ2)]

15) If $A = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 4 \end{bmatrix}$, then show that $|3A| = 27|A|$ [☞P 72(97)]

16) If $y = \sin^{-1}\left(\frac{2x}{1+x^2}\right)$ then find $\frac{dy}{dx}$ [☞P 116(235)]

17)* Find the maximum and minimum values of $f(x) = 9x^2 + 12x + 2$ [☞P 119(251)]

18)* Find $\int_0^{\pi/2} \sin^2 x \cos x \, dx$ [☞P 123(274)]

19) Find a particular solution of $\frac{dy}{dx} = y \tan x; y = 1$ when $x = 0$ [☞P 127(292)]

20) Find the angle between the vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$. [☞P 145(17)]

21) Find the direction cosines of a line which makes equal angles with the coordinate axes. [☞P 109(210)]

22) Evaluate $P(A \cup B)$, if $2P(A) = P(B) = \frac{5}{13}$ and $P(A|B) = \frac{2}{5}$ [☞P 113(224)]

SECTION-C

III. Answer ANY SEVEN the questions.

Each question carries FOUR mark. $7 \times 4 = 28$

23)* Let $f : N \to Y$ be a function defined as $f(x) = 4x + 3$, where, $Y = \{y \in N : y = 4x + 3 \text{ for some } x \in N\}$. Show that f is invertible. Find the inverse. [☞P 55(65)]

24) Prove that $\tan^{-1}\frac{63}{16} = \sin^{-1}\frac{5}{13} + \cos^{-1}\frac{3}{5}$ [☞P 58(71)]

25) Find x, if $\begin{bmatrix} x & -5 & -1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix} \begin{bmatrix} x \\ 4 \\ 1 \end{bmatrix} = O$ [☞P 101(176)]

26)* Find the values of k if area of triangle is 4 sq. units and vertices are (k, 0), (4, 0), (0, 2). [☞P 67(87)]

27) Differentiate the function $(\log x)^{\cos x}$ w.r.t. x. [☞P 82(124)]

Mahesh

28) Evaluate $\int \frac{1}{1+\tan x} dx$. [☞P 93(75)]

29) Find the area enclosed by the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ [☞P 92(151)]

30) Find a unit vector perpendicular to each of the vector $\vec{a}+\vec{b}$ and $\vec{a}-\vec{b}$, where $\vec{a}=3\hat{i}+2\hat{j}+2\hat{k}$ and $\vec{b}=\hat{i}+2\hat{j}-2\hat{k}$ [☞P 73(101)]

31) Minimise $Z=3x+2y$ subject to the constraints: $x+y\geq 8, 3x+5y\leq 15$; $x\geq 0, y\geq 0$ [☞P 78(115)]

32) A fair coin and an unbiased die are tossed. Let A be the event 'head appears on the coin' and B be the event '3 on the die'. Check whether A and B are independent events or not. [☞P 76(111)]

SECTION-D

IV. Answer ANY FIVE the questions. $5 \times 8 = 40$

Each question carries EIGHT marks.

33)* Solve the system of equations using matrix method [☞P 17(3)]

$x-y+2z=7, 3x+4y-5z=-5, 2x-y+3z=12$

34)* Find two positive numbers whose sum is 15 and the sum of whose squares is minimum. [☞P 31(27)]

35)* Find $\int \frac{x^2+x+1dx}{(x+2)(x^2+1)}$ [☞P 38(41)]

36) Evaluate $\int_0^1 \frac{\log(1+x)}{1+x^2} dx$ [☞P 36(37)]

37) Find the particular solution of the differential equation $\frac{dy}{dx}-3y\cot x=\sin 2x$; $y=2$ when $x=\frac{\pi}{2}$ [☞P 46(54)]

38)* If $\vec{a}=2\vec{i}+\vec{j}-3\vec{k}$, $\vec{b}=\vec{i}-2\vec{j}+\vec{k}, \vec{c}=-\vec{i}+\vec{j}-4\vec{k}$ and $\vec{d}=\vec{i}+\vec{j}+\vec{k}$ then compute $|(\vec{a}\times\vec{b})\times(\vec{c}+\vec{d})|$ [☞P 19(6)]

39) Find the shortest distance between the lines whose vector equations are $\vec{r}=(1-t)\hat{i}+(t-2)\hat{j}+(3-2t)\hat{k}$ and $\vec{r}=(s+1)\hat{i}+(2s-1)\hat{j}-(2s+1)\hat{k}$ [☞P 23(14)]

40)* A bag contains 4 red and 4 black balls, another bag contains 2 red and 6 black balls. One of the two bags is selected at random and a ball is drawn from the bag which is found to be red. Find the probability that the ball is drawn from the first bag. [☞P 26(20)]

Mahesh

GUESS PAPERS

Q3 GUESS PAPER -2

SR. MATHS

Time : 3 Hours Max.Marks : 100

SECTION - A

Ans-Page Index

I. Answer ALL the questions. Each question carries ONE mark. $12 \times 1 = 12$

1) The value of $\sin^{-1}\left(\cos\frac{3\pi}{5}\right)$ is [☞P 132(10)]

1) $\frac{\pi}{10}$ 2) $\frac{3\pi}{5}$ 3) $-\frac{\pi}{10}$ 4) $-\frac{3\pi}{5}$

2) If $\begin{vmatrix} x & 2 \\ 18 & x \end{vmatrix} = \begin{vmatrix} 6 & 2 \\ 18 & 6 \end{vmatrix}$, then x is equal to [☞P 134(1)]

1) 6 2) ± 6 3) − 6 4) 0

3) The maximum value of $\frac{\log x}{x}, 0 < x < \infty$ is [☞P 136(12)]

1) ∞ 2) e 3) 1 4) e^{-1}

4) The area bounded by the curve $y=x^2-4$, and the lines y=0 and y=5 is [☞P 138(5)]

1) $\frac{38}{3}$ 2) $\frac{76}{3}$ 3) $\frac{16}{3}$ 4) $\frac{8}{3}$

5) The degree of the differential equation $\left(\frac{d^2y}{dx^2}\right)^3 + \left(\frac{dy}{dx}\right)^2 + \sin\left(\frac{dy}{dx}\right) + 1 = 0$ is [☞P 139(12)]

1) 3 2) 2 3) 1 4) not defined

6)* Which of the following is not a component of linear programming problem? [☞P 141(9)]

1) Objective function 2) Constriant

3) Decision variable 4) Differential equation

7)* Define an equivalence relation [☞P 97(162)]

8) Find X, if $Y=\begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$ and $2X+Y=\begin{bmatrix} 1 & 0 \\ -3 & 2 \end{bmatrix}$ [☞P 98(164)]

9) Differentiate $\sin(x^2+5)$ with respect to x. [☞P 115(230)]

10) Find. $\int \frac{(\log x)^2}{x} dx$ [☞P 121(262)]

11) Find the value of x for which $x(\hat{i}+\hat{j}+\hat{k})$ is a unit vector. [☞P 106(195)]

12)* If $P(A)=\frac{3}{5}$ and $P(B)=\frac{1}{5}$, find P (A ∩ B) if A and B are independent events. [☞P 112(221)]

☞ **Guess Paper 2** is unofficial BMP2 in which * Problems are modified

Mahesh

SECTION-B

II. Answer ALL the questions. $10 \times 2 = 20$

Each question carries TWO marks.

13)* Prove that the function $f : R \to R$, given by $f(x) = 2x$, is one-one and onto. [☞P 96(154)]

14)* Compute $\begin{bmatrix} a^2+b^2 & b^2+c^2 \\ a^2+c^2 & a^2+b^2 \end{bmatrix} + \begin{bmatrix} 2ab & 2bc \\ -2ac & -2ab \end{bmatrix}$ [☞P 102(180)]

15) Find area of the triangle with vertices (1, 0), (6, 0), (4, 3) using determinants. [☞P 67(86)]

16) Find $\frac{dy}{dx}$, if $x = a(\theta + \sin\theta)$, $y = a(1 - \cos\theta)$. [☞P 117(240)]

17) Find the intervals in which the function f given by $f(x) = 2x^3 - 3x^2 - 36x + 7$ is increasing [☞P 119(PQ)]

18) Evaluate $\int_0^4 |x-1|\, dx$ [☞P 124(273)]

19) Find the general solution of $\frac{dy}{dx} + \frac{y}{x} = x^2$ [☞P 127(289)]

20) Find $\vec{a} \times \vec{b}$, if $\vec{a} = 2\hat{i}+\hat{j}+3\hat{k}$ and $\vec{b} = 3\hat{i}+5\hat{j}-2\hat{k}$ [☞P 108(206)]

21) If a line makes angles 90°, 60° and 30° with the positive x,y and z-axes respectively, find its direction cosines. [☞P 109(PQ)]

22) A family has two children. What is the probability that both the children are boys given that at least one of them is a boy ? [☞P 114(228)]

SECTION-C

III. Answer ANY SEVEN the questions.

Each question carries FOUR mark. $7 \times 4 = 28$

23)* State whether the function $f : R \to R$ defined by $f(x) = 3 - 4x$ is one-one, onto or bijective. Justify your answer. [☞P 55(64)]

24) Solve $\arcsin\left(\frac{5}{x}\right) + \arcsin\frac{12}{x} = \frac{\pi}{2}$ $(x>0)$ [☞P 59(75)]

25)* **If** $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{bmatrix}$, **then show that** $A^3 - 23A - 40I = O$ [☞P 65(84)]

26) Find the inverse of the matrix $\begin{bmatrix} 2 & 1 & 3 \\ 4 & -1 & 0 \\ -7 & 2 & 1 \end{bmatrix}$ if it exists. [☞P 69(93)]

27) Find the relationship between a and b so that the function f defined by $f(x) = \begin{cases} ax+1, & \text{if } x \le 3 \\ bx+3, & \text{if } x > 3 \end{cases}$ is continuous at $x = 3$. [☞P 80(118)]

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Ans-Page Index

28)* Evaluate $\int_0^a \frac{\sqrt{x}}{\sqrt{x}+\sqrt{a-x}}dx$ [☞P 84(130)]

29) Find the area bounded by the curve $y = \cos x$ between $x = 0$ and $x = 2\pi$. [☞P 91(48)]

30) Find a unit vector perpendicular to each of the vectors $(\vec{a}+\vec{b})$ and $(\vec{a}-\vec{b})$ where $\vec{a}=\hat{i}+\hat{j}+\hat{k}$, $\vec{b}=\hat{i}+2\hat{j}+3\hat{k}$ [☞P 73(PQ)]

31)* Solve the Linear Programming Problem graphically: [☞P 79(116)]
Minimise $Z = -3x + 4y$ subject to $x + 2y \le 8$, $3x + 2y \le 12$, $x \ge 0$, $y \ge 0$.

32) Given that the two numbers appearing on throwing two dice are different. [☞P 76(110)]
Find the probability of the event 'the sum of numbers on the dice is 4'.

SECTION-D

IV. Answer ANY FIVE the questions.

Each question carries EIGHT marks. $5 \times 8 = 40$

33) Solve system of $x - y + z = 4$, $2x + y - 3z = 0$, $x + y + z = 2$ using matrix method. [☞P 18(PQ)]

34) Show that the semi-vertical angle of the cone of the maximum volume and of given slant height is $\tan^{-1}\sqrt{2}$ [☞P 34(33)]

35) Find $\int \frac{x^4 dx}{(x-1)(x^2+1)}$ [☞P 39(42)]

36)* Evaluate $\int_0^{\frac{\pi}{2}} \log \sin x dx$ [☞P 37(39)]

37) Show that the differential equation $(x^2 - y^2)\,dx + 2xy\,dy = 0$ is homogeneous and solve it. [☞P 48(56)]

38) If $\bar{a}=\bar{i}-2\bar{j}+3\bar{k}$, $\bar{b}=2\bar{i}+\bar{j}+\bar{k}$, $\bar{c}=\bar{i}+\bar{j}+2\bar{k}$ then find $|(\bar{a}\times\bar{b})\times\bar{c}|$ and $|\bar{a}\times(\bar{b}\times\bar{c})|$ [☞P 21(9)]

39) Find the shortest distance between the lines [☞P 24(15)]

$$\frac{x+1}{7}=\frac{y+1}{-6}=\frac{z+1}{1} \quad \text{and} \quad \frac{x-3}{1}=\frac{y-5}{-2}=\frac{z-7}{1}$$

40) In a factory which manufactures bolts, machines A, B and C manufacture respectively 25%, 35% and 40% of the bolts. Of their outputs, 5, 4 and 2 percent are respectively defective bolts. A bolt is drawn at random from the product and is found to be defective. What is the probability that it is manufactured by the machine B? [☞P 27(22)]

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GUESS PAPERS

Q 4 GUESS PAPER -3

SR. MATHS

Time : 3 Hours Max.Marks : 100

Ans-Page Index

SECTION-A

I. Answer ALL the questions. $12 \times 1 = 12$

Each question carries ONE mark.

1. $\tan^{-1}\sqrt{3} - \cot^{-1}(-\sqrt{3})$ is equal to [☞P 132(5)]

1) π 2) $-\frac{\pi}{2}$ 3) 0 4) $2\sqrt{3}$

2. If area of triangle is 35 sq units with vertices $(2, -6)$, $(5, 4)$ and $(k, 4)$. Then k is [☞P 134(6)]

1) 12 2) –2 3) –12, –2 4) 12, –2

3. For all real values of x, the minimum value of $\frac{1-x+x^2}{1+x+x^2}$ is [☞P 136(6)]

1) 0 2) 1 3) 3 4) 1/3

4. Area bounded by the curve $y = x^3$, the x-axis and the ordinates $x = -2$ and $x = 1$ is [☞P 138(2)]

1) -9 2) $\frac{-15}{4}$ 3) $\frac{15}{4}$ 4) $\frac{17}{4}$

5. If $\vec{a}+\vec{b}+\vec{c}=\vec{0}$, $|\vec{a}|=3$, $|\vec{b}|=5$, $|\vec{c}|=7$ then the angle between $\vec{a}$ and $\vec{b}$ is [☞P 140(12

1) $\frac{\pi}{6}$ 2) $\frac{2\pi}{3}$ 3) $\frac{5\pi}{3}$ 4) $\frac{\pi}{3}$

6. When two dice are rolled, the probability of getting unequal numbers on the faces is [☞P 142(13

1) $\frac{1}{6}$ 2) $\frac{35}{36}$ 3) $\frac{5}{6}$ 4) $\frac{1}{3}$

7. Define symmetric relation [☞P 97(160)]

8. Find the values of x for which $\begin{vmatrix} 3 & x \\ x & 1 \end{vmatrix} = \begin{vmatrix} 3 & 2 \\ 4 & 1 \end{vmatrix}$ [☞P 103(183

9. Show that the function given by $f(x) = e^{2x}$ is increasing on R. [☞P 118(245

10. Find the area enclosed between the curve $y = x^2$, X-axis and the lines $x = -1, x = 2$ [☞P 90(143)]

11. Find the general solution of the differential equation $\frac{dy}{dx} = \frac{1+y^2}{1+x^2}$ [☞P 127(287

12. Define Optimal Solution [☞P 145(22

Mahesh

SECTION-B

II. Answer ALL the questions.
Each question carries TWO marks. $10 \times 2 = 20$

13. Check the injectivity and surjectivity of the function $f : Z \to Z$ given by $f(x) = x^2$. [☞P 96(157)]

14. Show that the matrix $A=\begin{bmatrix} 0 & 1 & -1 \\ -1 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ is a skew symmetric matrix. [☞P 102(178)]

15. If $A=\begin{bmatrix} 1 & 2 \\ 4 & 2 \end{bmatrix}$, then show that $|2A|=4|A|$ [☞P 103(185)]

16. Find $\frac{dy}{dx}$, if $x = at^2$, $y = 2at$. [☞P 116(238)]

17. Find the intervals in which the function $x^2 + 2x - 5$ is strictly increasing or decreasing: [☞P 118(249)]

18. Evaluate $\int_0^4 |x-1|\,dx$ [☞P 124(275)]

19. Find the general solution of $\frac{dy}{dx}+\frac{y}{x}=x^2$ [☞P 127(291)]

20. Find the area of a parallelogram whose adjacent sides are given by the vectors $\vec{a} = 3\hat{i} + \hat{j} + 4\hat{k}$ and $\vec{b} = \hat{i} - \hat{j} + \hat{k}$ [☞P 108(207)]

21. Show that the points (2, 3, 4), (– 1, – 2, 1), (5, 8, 7) are collinear. [☞P 110(212)]

22. Two cards are drawn at random in succession and without replacement from a pack of 52 playing cards. Find the probability that both the cards are black. [☞P 114(227)]

SECTION-C

III. Answer ANY SEVEN the questions.
Each question carries FOUR mark. $7 \times 4 = 28$

23. Let $f : \{2, 3, 4, 5\} \to \{3, 4, 5, 9\}$ and $g : \{3, 4, 5, 9\} \to \{7, 11, 15\}$ be functions defined as $f(2) = 3$, $f(3) = 4$, $f(4) = f(5) = 5$ and $g(3) = g(4) = 7$ and $g(5) = g(9) = 11$. Find gof. [☞P 93(5)]

24. Prove that $\sin^{-1}\left(\frac{3}{5}\right)+\cos^{-1}\left(\frac{12}{13}\right)=\cos^{-1}\left(\frac{33}{65}\right)$ [☞P 57(PQ)]

25. If $A=\begin{bmatrix} 0 & 6 & 7 \\ -6 & 0 & 8 \\ 7 & -8 & 0 \end{bmatrix}, B=\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}, C=\begin{bmatrix} 2 \\ -2 \\ 3 \end{bmatrix}$ [☞P 63(79)]

Calculate AC, BC and (A + B)C. Also, verify that (A + B)C = AC + BC

26. Show that $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}^2 = (a^3+b^3+c^3-3abc)^2$ [☞P 72(100)]

27. Differentiate $x^{\sin x}$, $x > 0$ w.r.t. x. [☞P 82(123)]

Manesh

28. Evaluate $\int_0^{\frac{\pi}{2}} \frac{\sqrt{\sin x}}{\sqrt{\sin x}+\sqrt{\cos x}} dx$.

29. Find the area between curve $y = x^2$ and the line $y = 2x$.

30. If $\vec{a} = \hat{i}+\hat{j}+\hat{k}$, $\vec{b} = 2\hat{i}-\hat{j}+3\hat{k}$ and $\vec{c} = \hat{i}-2\hat{j}+\hat{k}$ find a unit vector parallel to the vector $2\vec{a}-\vec{b}+3\vec{c}$.

31. Maximise $Z = 3x + 2y$ subject to $x + 2y \le 10$, $3x + y \le 15$, $x, y \ge 0$.

32. Let A and B be independent events with $P(A) = 0.3$ and $P(B) = 0.4$. Find
(i) $P(A \cap B)$ (ii) $P(A \cup B)$ (iii) $P(A|B)$ (iv) $P(B|A)$

SECTION-D

IV. Answer ANY FIVE the questions. $5 \times 8 = 40$
Each question carries EIGHT marks.

33. Solve the following system of equations by matrix method.
$3x - 2y + 3z = 8$, $2x + y - z = 1$, $4x - 3y + 2z = 4$

34. An open topped box is to be constructed by removing equal squares from each corner of a 3 metre by 8 metre rectangular sheet of aluminium and folding up the sides. Find the volume of the largest such box.

35. Find $\int \frac{x^2}{(x^2+1)(x^2+4)} dx$.

36. Evaluate $\int_0^{\pi/4} \frac{\sin x+\cos x}{9+16\sin 2x} dx$.

37. Solve the differential equation $(\tan^{-1} y - x)\, dy = (1 + y^2)\, dx$.

38. If $\bar{a} = 7\bar{i}-2\bar{j}+3\bar{k}, \bar{b} = 2\bar{i}+8\bar{k}, \bar{c} = \bar{i}+\bar{j}+\bar{k}$, then compute $\bar{a}\times\bar{b}, \bar{a}\times\bar{c}, \bar{a}\times(\bar{b}+\bar{c})$.
Verify whether cross product is distributive over vector addition.

39. Find the shortest distance between the lines whose vector equations are
$\vec{r} = (\hat{i}+2\hat{j}+3\hat{k})+\lambda(\hat{i}-3\hat{j}+2\hat{k})$ and $\vec{r} = 4\hat{i}+5\hat{j}+6\hat{k}+\mu(2\hat{i}+3\hat{j}+\hat{k})$

40. Coloured balls are distributed in four boxes as shown in the following table:

Box	Colour			
	Black	White	Red	Blue
I	3	4	5	6
II	2	2	2	2
III	1	2	3	1
IV	4	3	1	5

A box is selected at random and then a ball is randomly drawn from the selected box. The colour of the ball is black, what is the probability that ball drawn is from the box III?

☞ P 83(127) ☞ P 90(146) ☞ P 73(102) ☞ P 78(115) ☞ P 75(107) ☞ P 16(1) ☞ P 33(PQ) ☞ P 39(43) ☞ P 37(38) ☞ P 44(49) ☞ P 20(8) ☞ P 22(12) ☞ P 28(23)

Mahesh